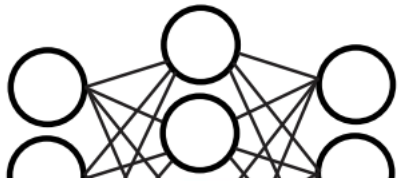
MEN. WOMEN. BABIES. ELDERLY.
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Self-Supervision II









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Reminder – Self supervised learning

Supervised Learning



Labeled data

Supervised Training

Feature extraction

Specific Task

Self-Supervised Learning



Unlabeled (massive) data

Self-Supervised Training (Pretext task)

Feature extraction

Supervised Fine Tuning

Specific Task

Representation Learning (Learning embeddings of data)

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Topics

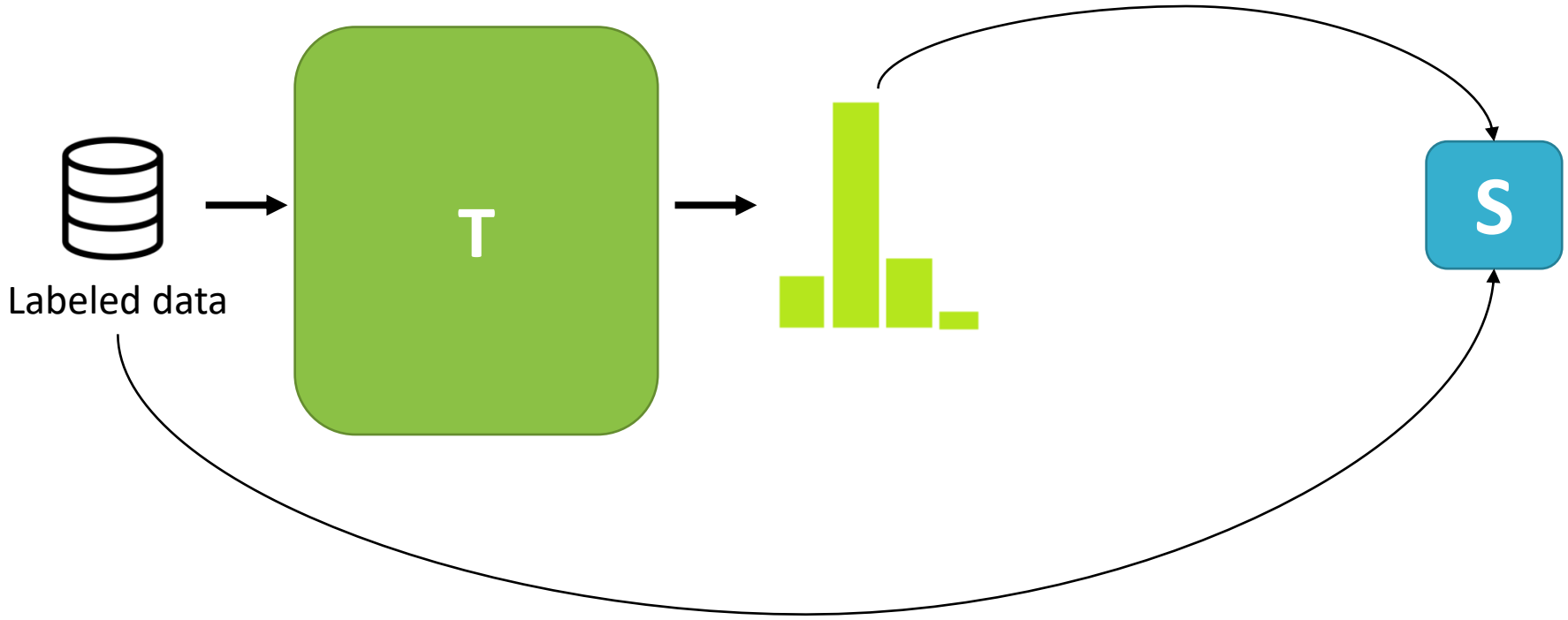
- self-DIstillation with NO labels
  - **DINO**
- Masked Auto Encoders
  - **MAE**
- Contrastive Language Image Pretraining
  - **CLIP**



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DiNO - Approach

- Self supervised learning as a special case of **knowledge distillation**





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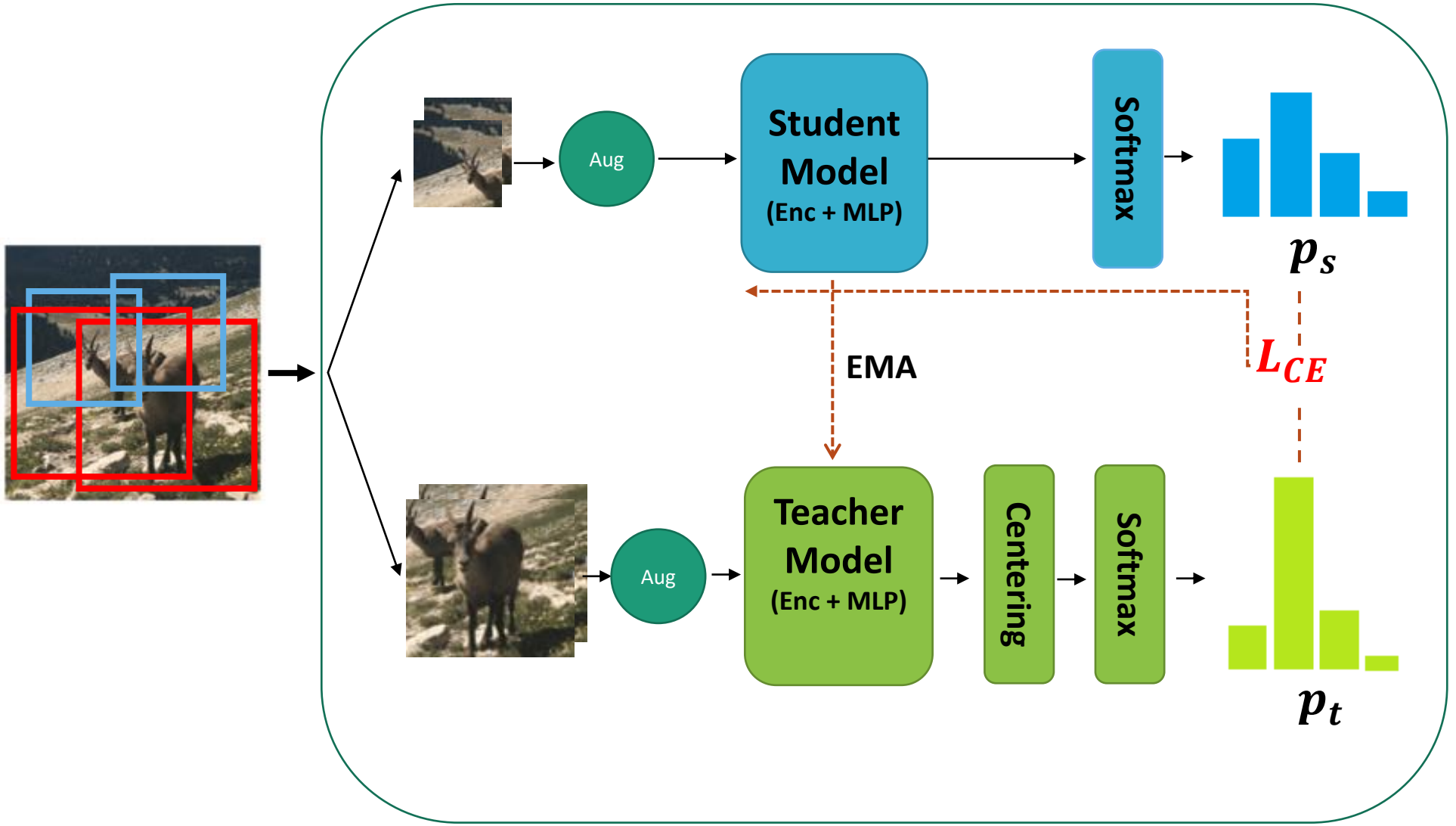
DiNO - Training



DiNO

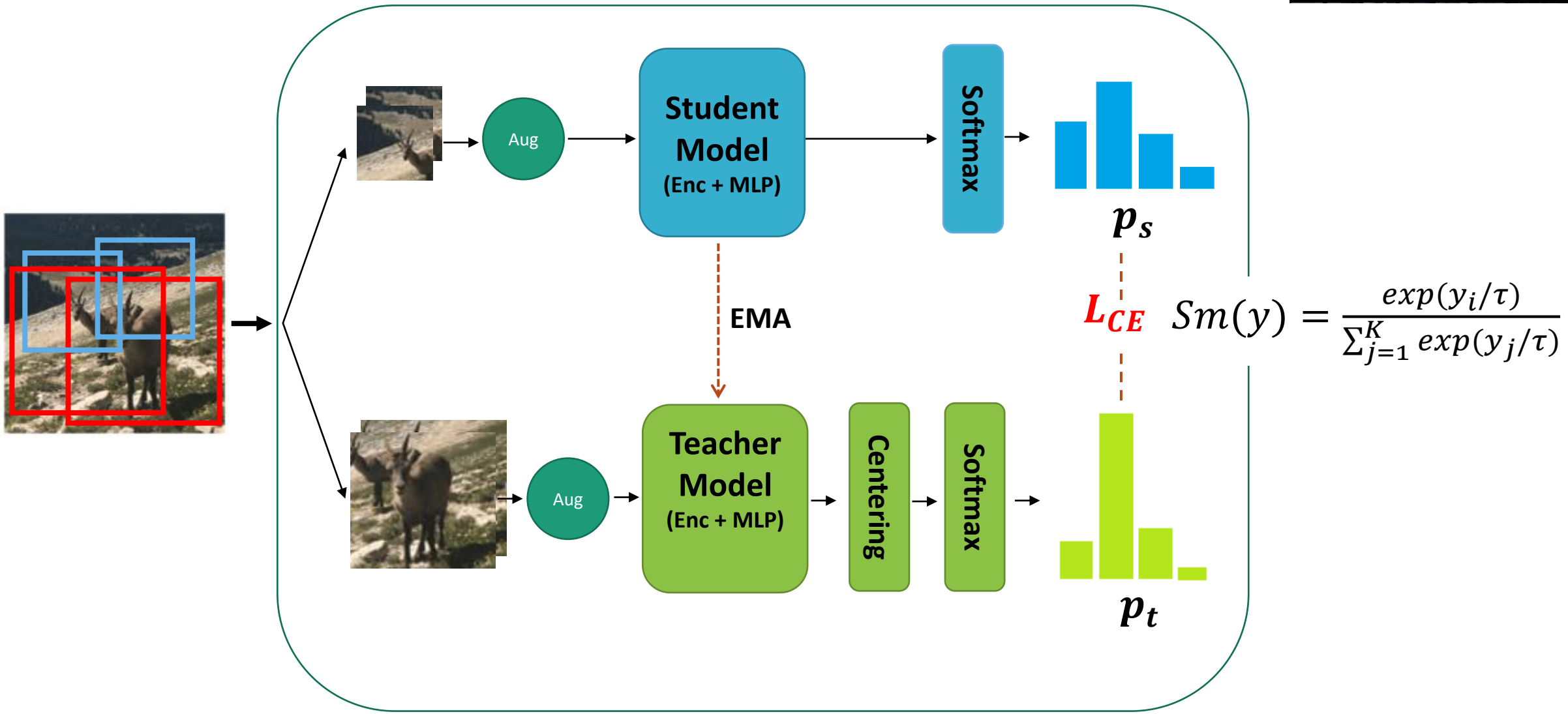
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DiNO - Training



MEN. WOMEN. BABIES. ELDERLY.
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DiNO - Training



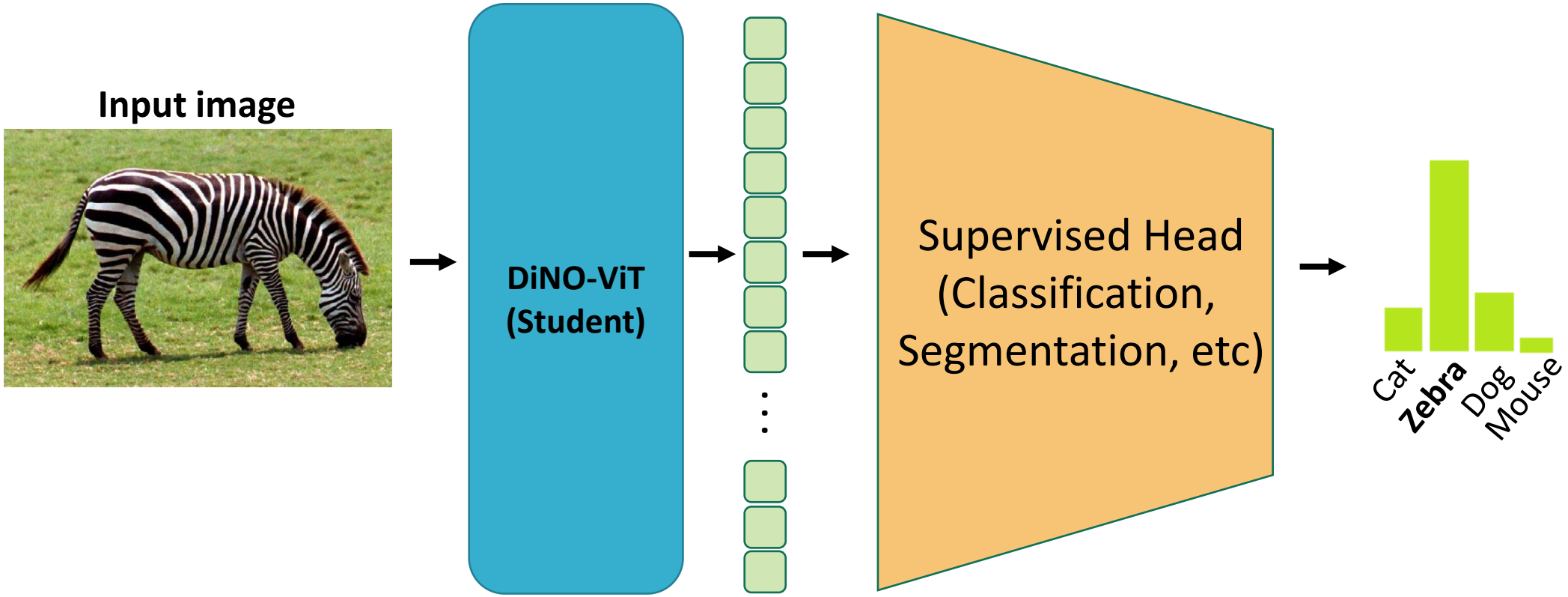
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DiNO - Explanation

- Augmentations
  - Tells the model what to ignore
    - Collor jitter, Gaussian Blur, Solarize
    - Acts as a data prior
- “Global – local” cropping
- Teacher out-distribution sharpening via centering & Low-temperature in softmax
- The student encoder learns “abstract representations”
  - No awareness of “class labels” or meaning behind logits

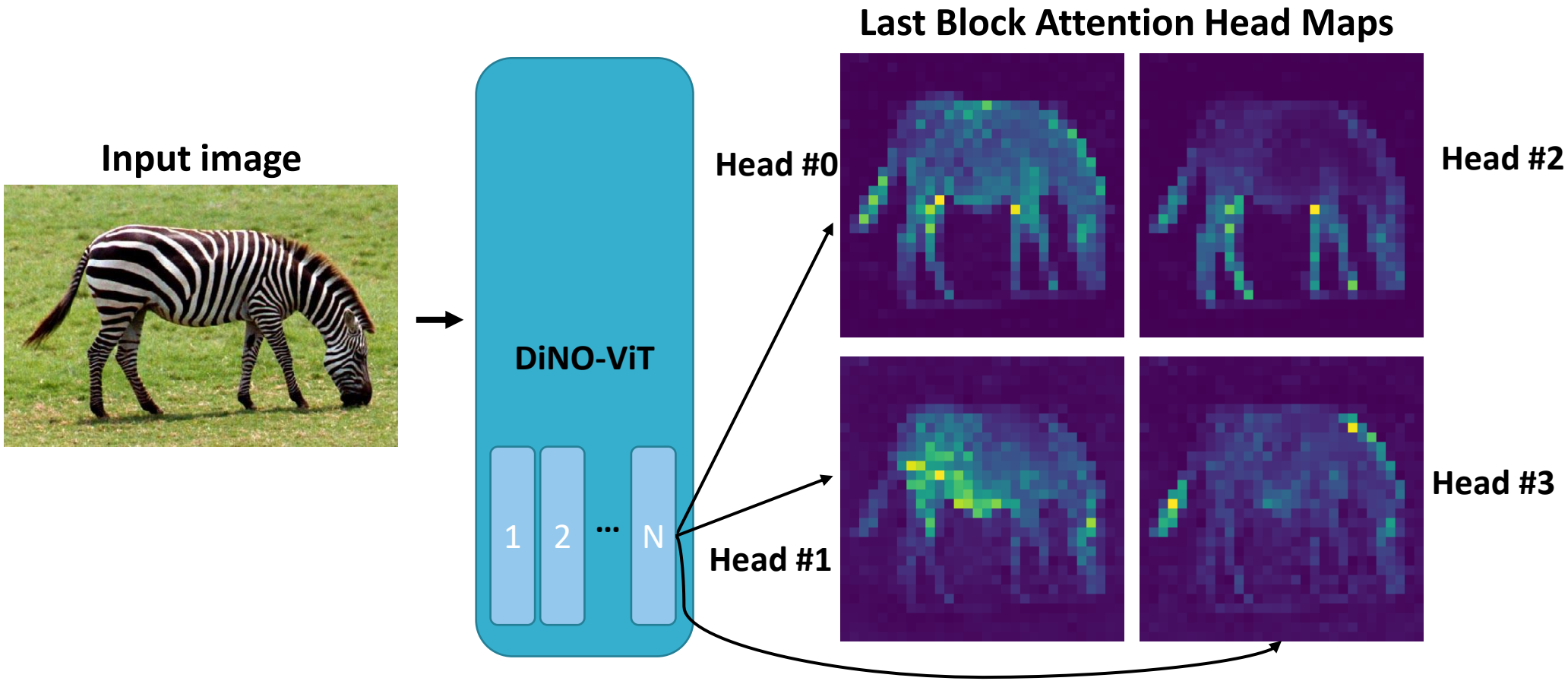
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DiNO - Inference #1



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DiNO - Inference #2



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DiNO - Inference #2

Input **Supervised Segmentation** **DiNO (SSL) Segmentation**

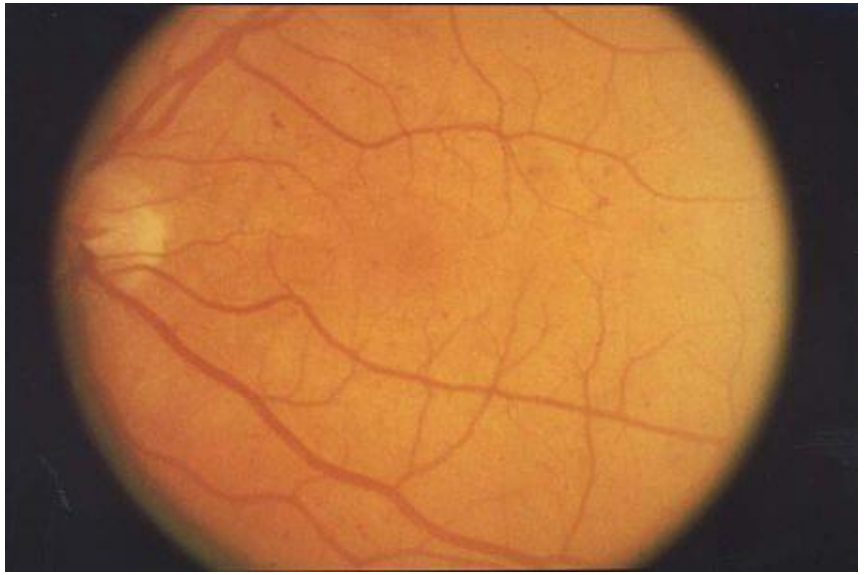


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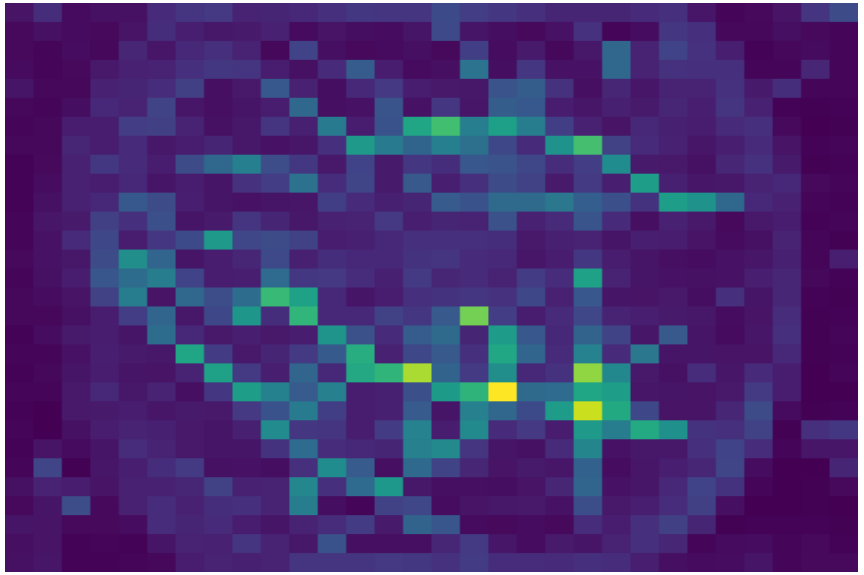
DiNO - Inference #2

- Self supervised learning also makes learned representations applicable to out-of-distribution data

Input image



Last block attention map



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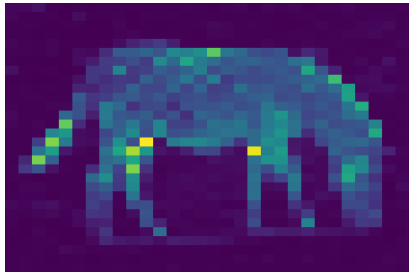
Usage

```
vit_model = torch.hub.load('facebookresearch/dino:main',
                           f'dino_vits16', pretrained=True)
img = imread('zebra.png')

x = vit_model.prepare_tokens(img)
for blk in vit_model.blocks[:-1]:
    x = blk(x)
attn_maps = vit_model.blocks[-1](x, return_attention=True)

# Choose head, Get attention map of class token
attn_map = attn_maps[0, HEAD, 0, 1:].reshape((1, 1, H_PATCHES, W_PATCHES))
attn_map = F.interpolate(attn_map, scale_factor=16, mode="nearest")
```





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Usage

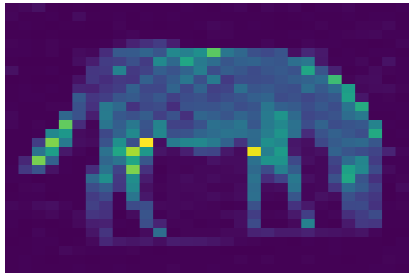
```
vit_model = torch.hub.load('facebookresearch/dino:main',
                           f'dino_vits16', pretrained=True)

img = imread('zebra.png')

attn_maps = vit_model.get_last_selfattention(img)

# Choose head, Get attention map of class token
attn_map = attn_maps[0, HEAD, 0, 1:].reshape((1, 1, H_PATCHES, W_PATCHES))
attn_map = F.interpolate(attn_map, scale_factor=16, mode="nearest")
```





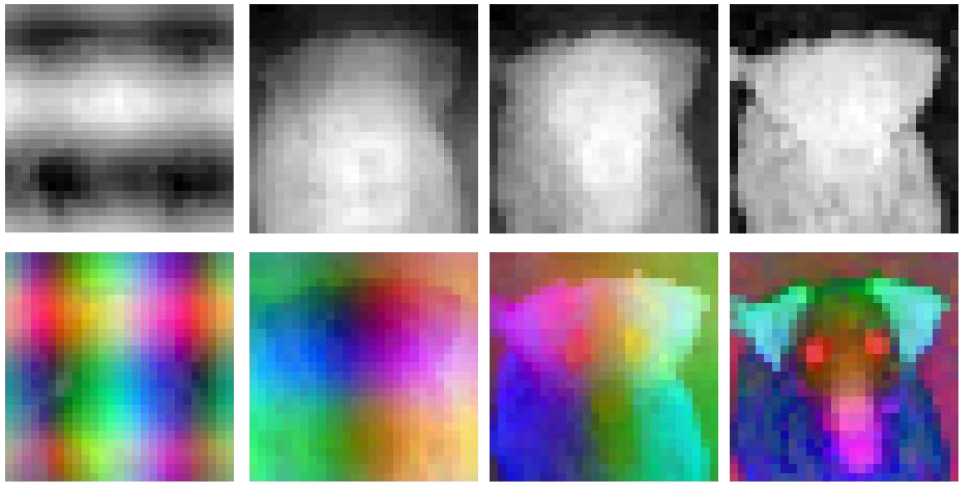
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PCA (Keys) across layers

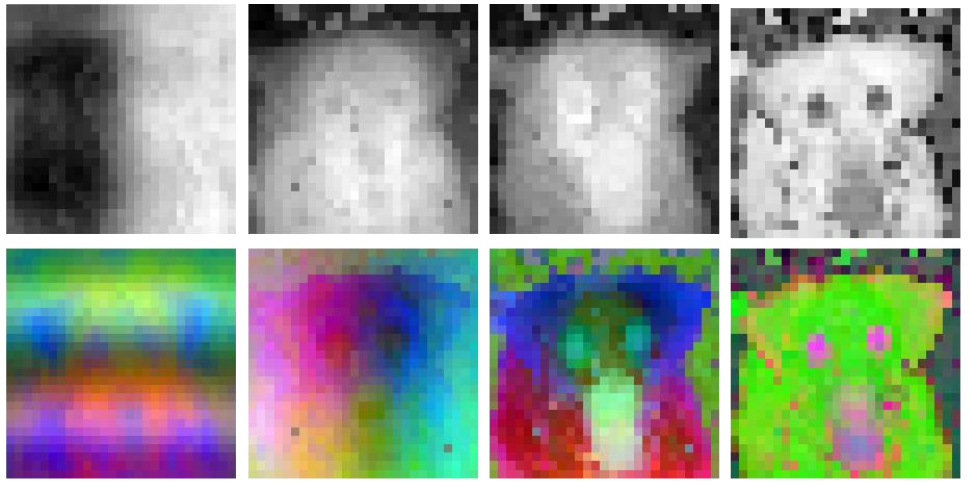


Self-supervised ViT (DINO-ViT)

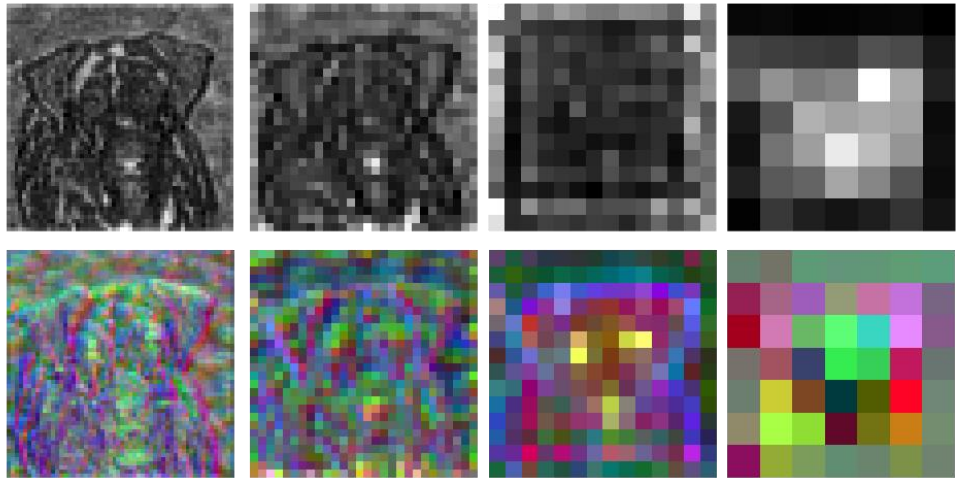


shallow deep

Supervised ViT

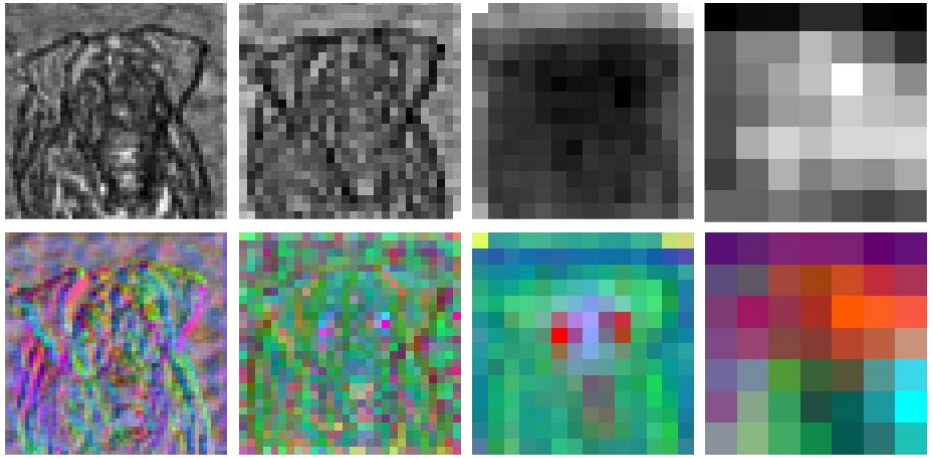


Self-supervised ResNet (DINO-ResNet)



shallow **(b) ResNet** deep

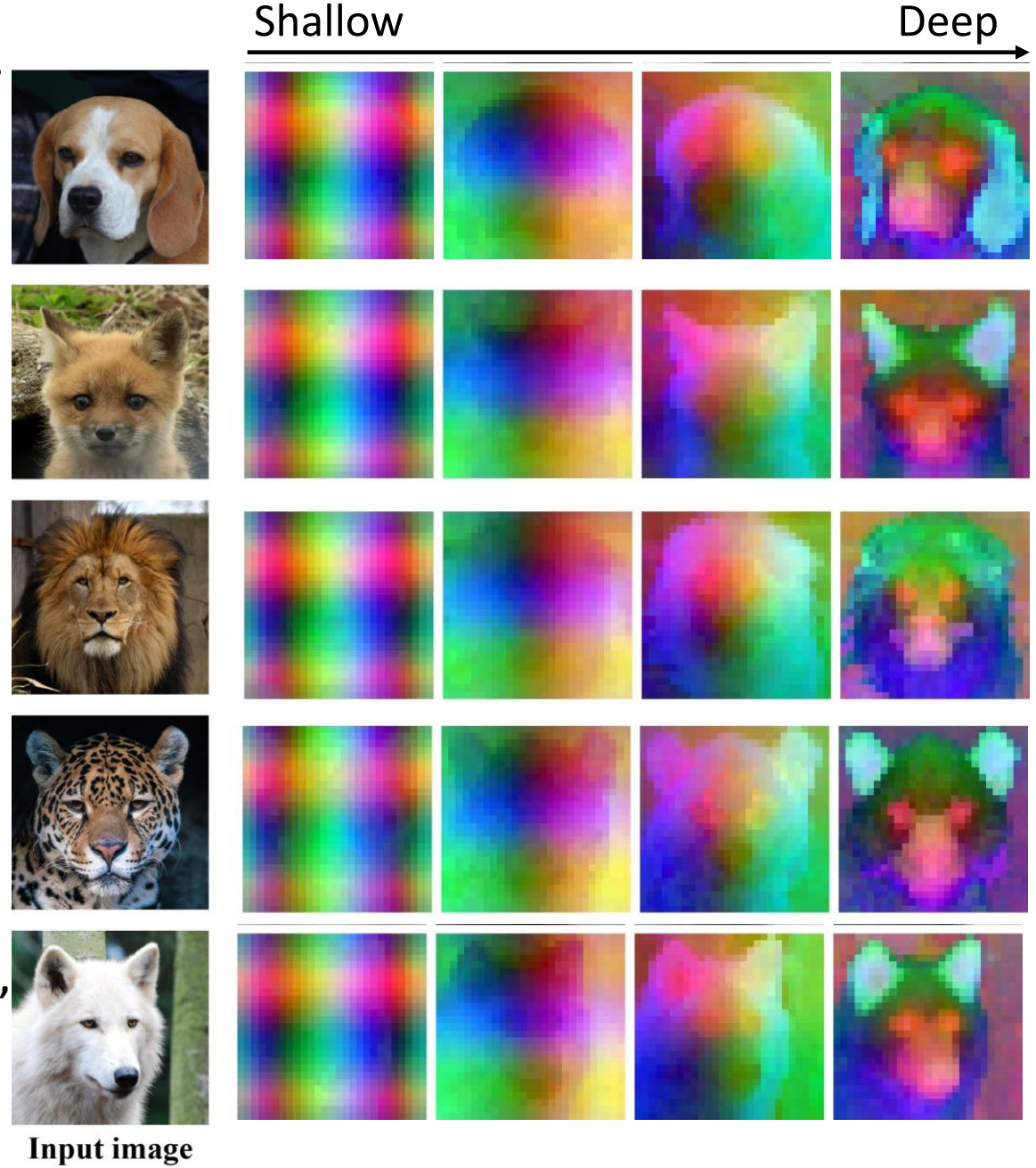
Supervised ResNet



WAIC

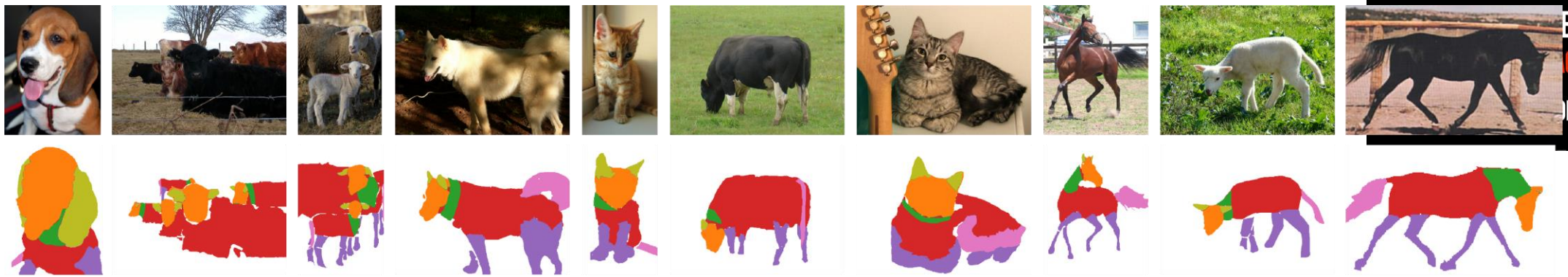
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PCA – DiNO ViT



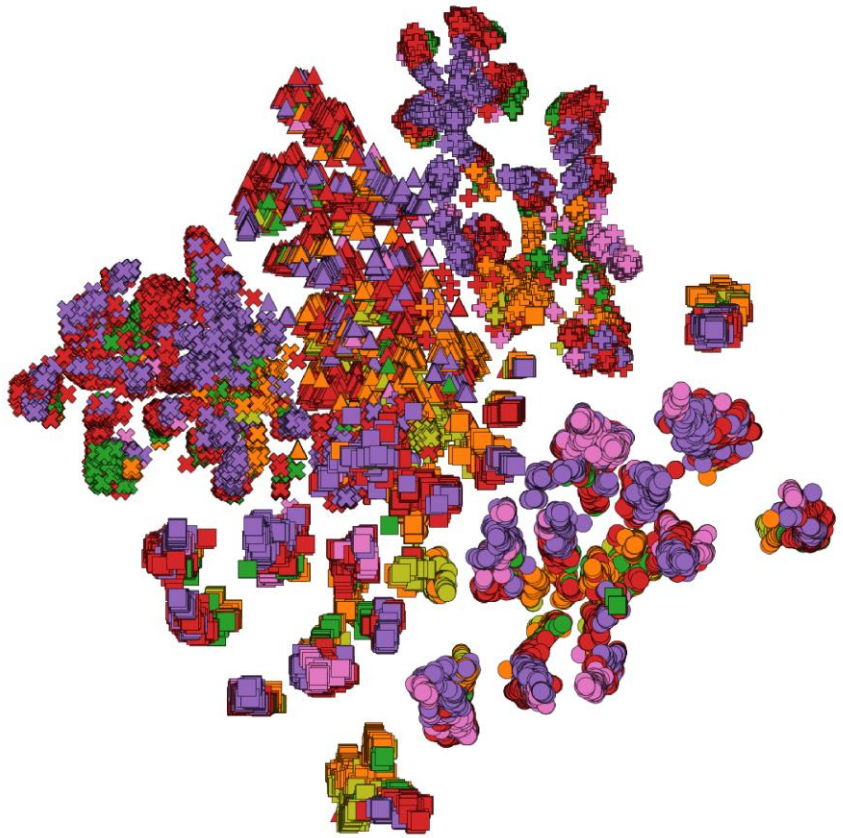
Amir, S., Gandelsman, Y., Bagon, S., Dekel, T.,
“Deep ViT Features as dense visual descriptors”

WAIC

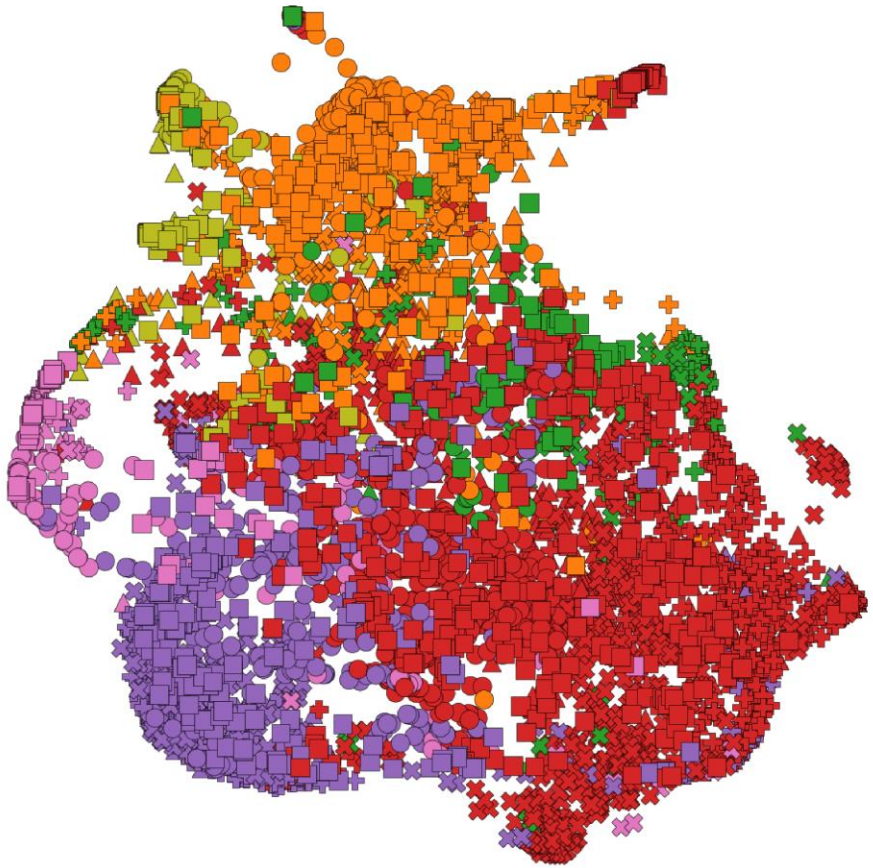


(a) Sample images and ground truth parts

Torso
Neck
Head
Ears
Tail
Limbs
○ Cat
□ Dog
✚ Horse
✡ Sheep
△ Cow



(c) Supervised ViT



(b) Self-Supervised ViT (DINO-ViT)

WAIC

[CLS] as a **global appearance** representation

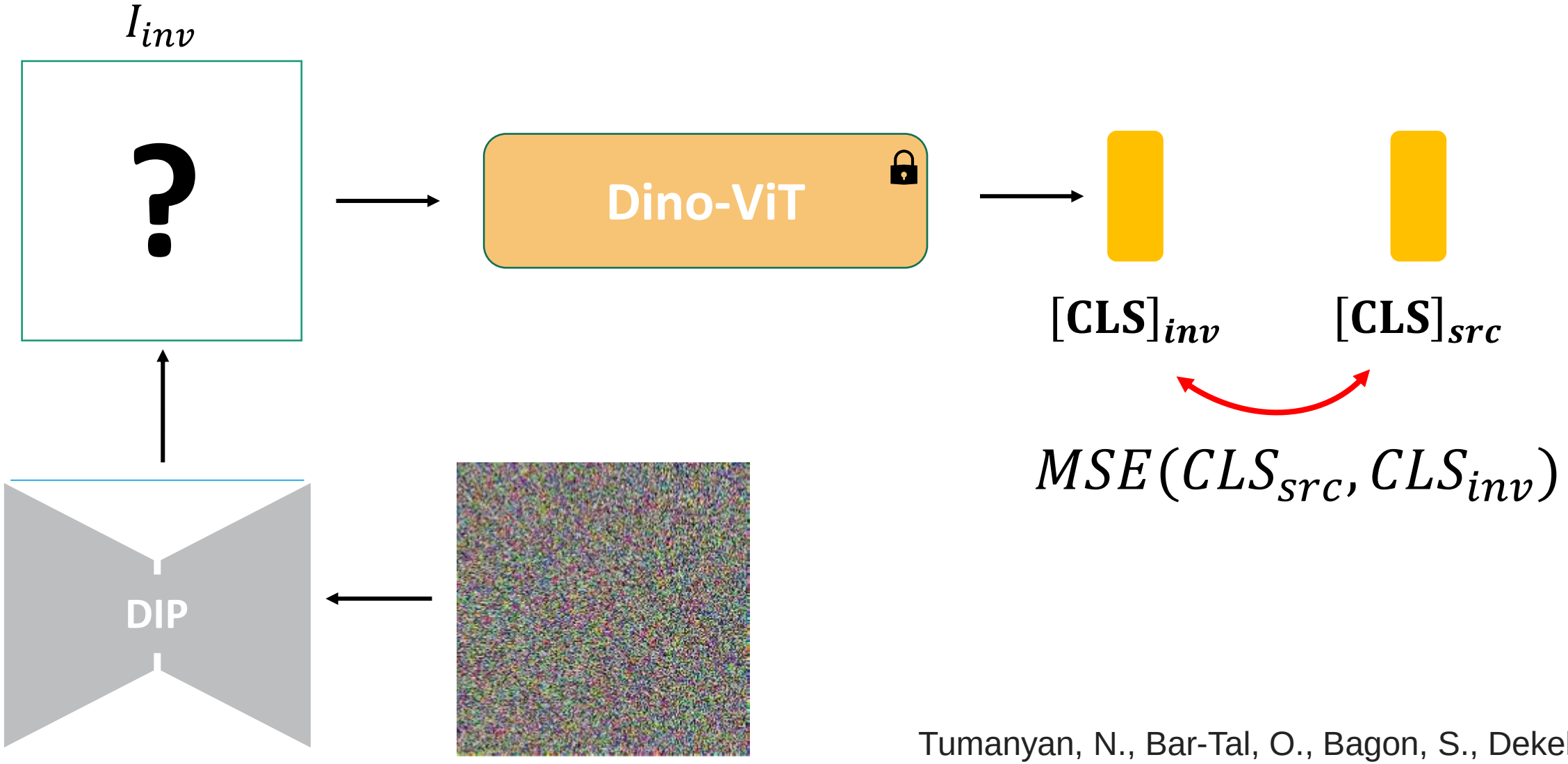
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Tumanyan, N., Bar-Tal, O., Bagon, S., Dekel, T.
Splicing vit features for semantic appearance transfer (CVPR 2022)

[CLS] as a **global appearance** representation

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Tumanyan, N., Bar-Tal, O., Bagon, S., Dekel, T.
Splicing vit features for semantic appearance transfer (CVPR 2022)

WAIC

[CLS] as a **global appearance** representation

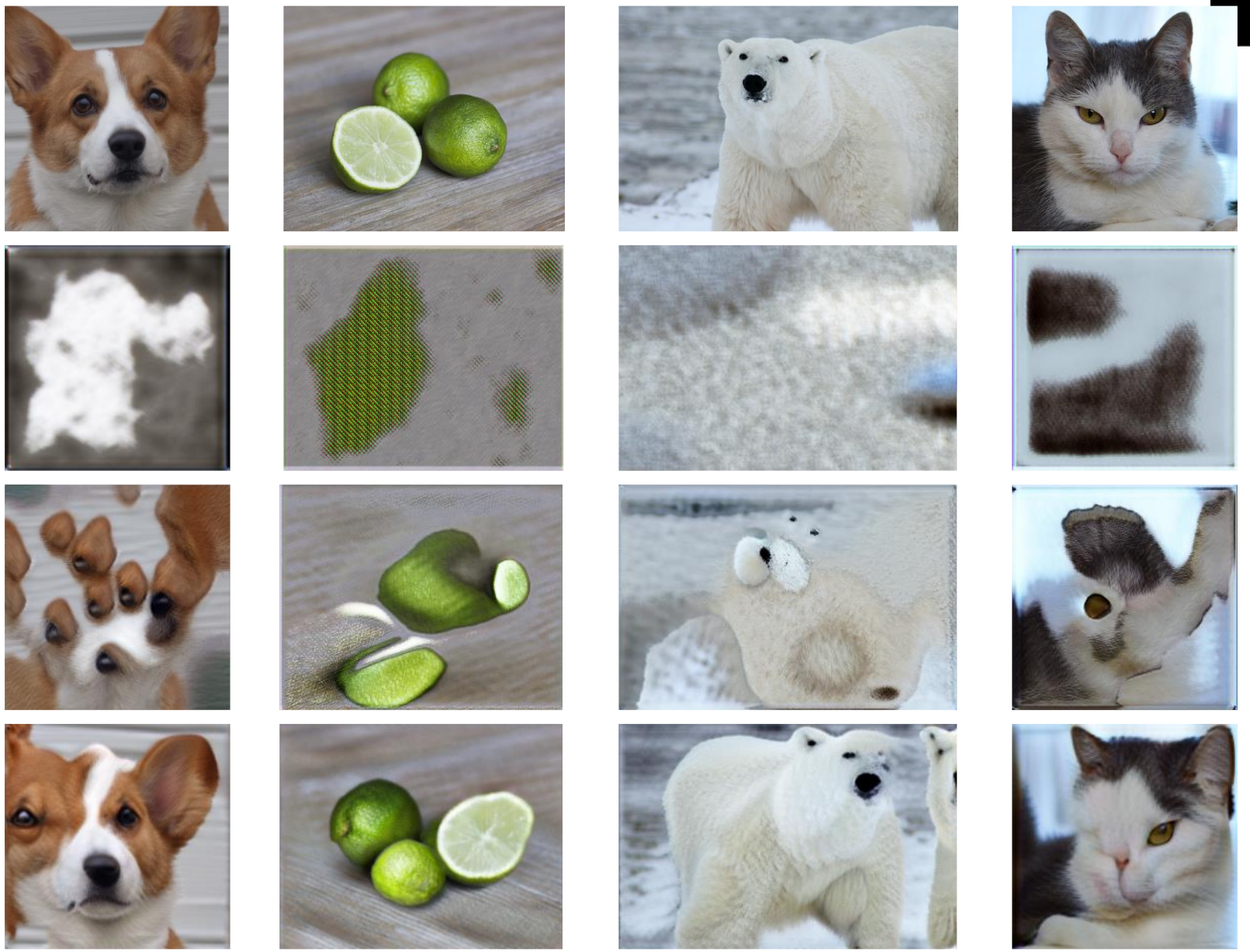
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Input

layer **0**

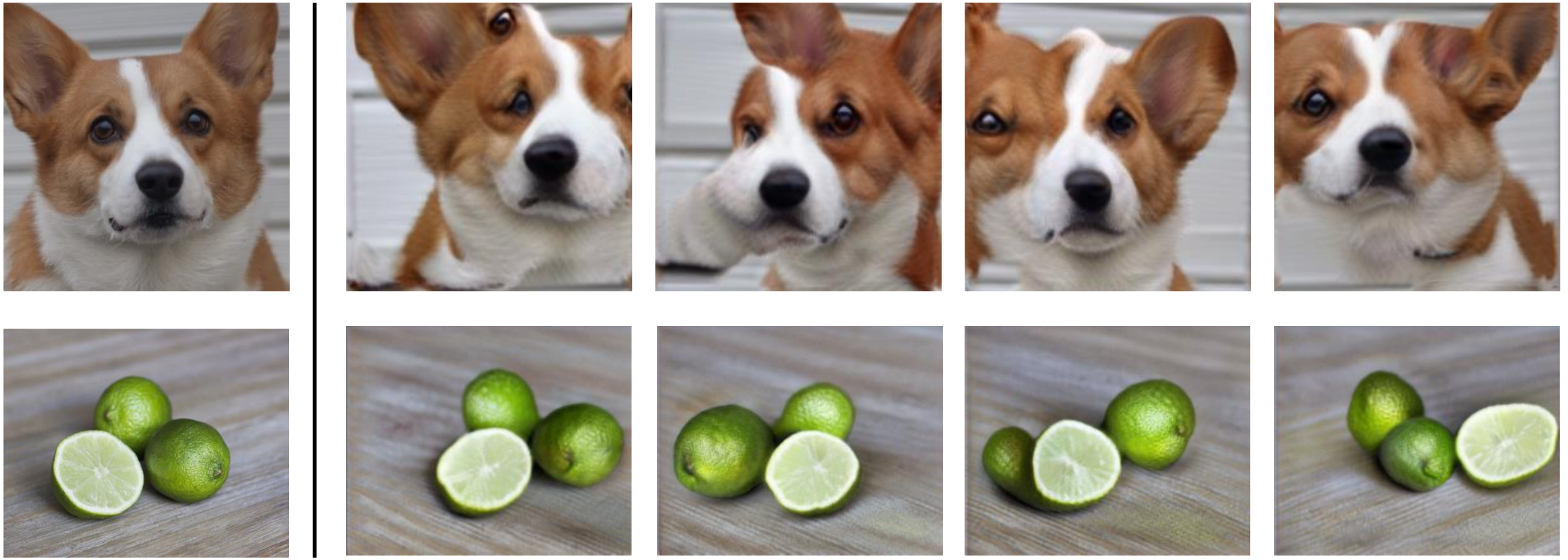
layer **3**

layer **11**



[CLS] as a **global appearance** representation

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Inversion run 1 | Inversion run 2 | Inversion run 3 | Inversion run 4

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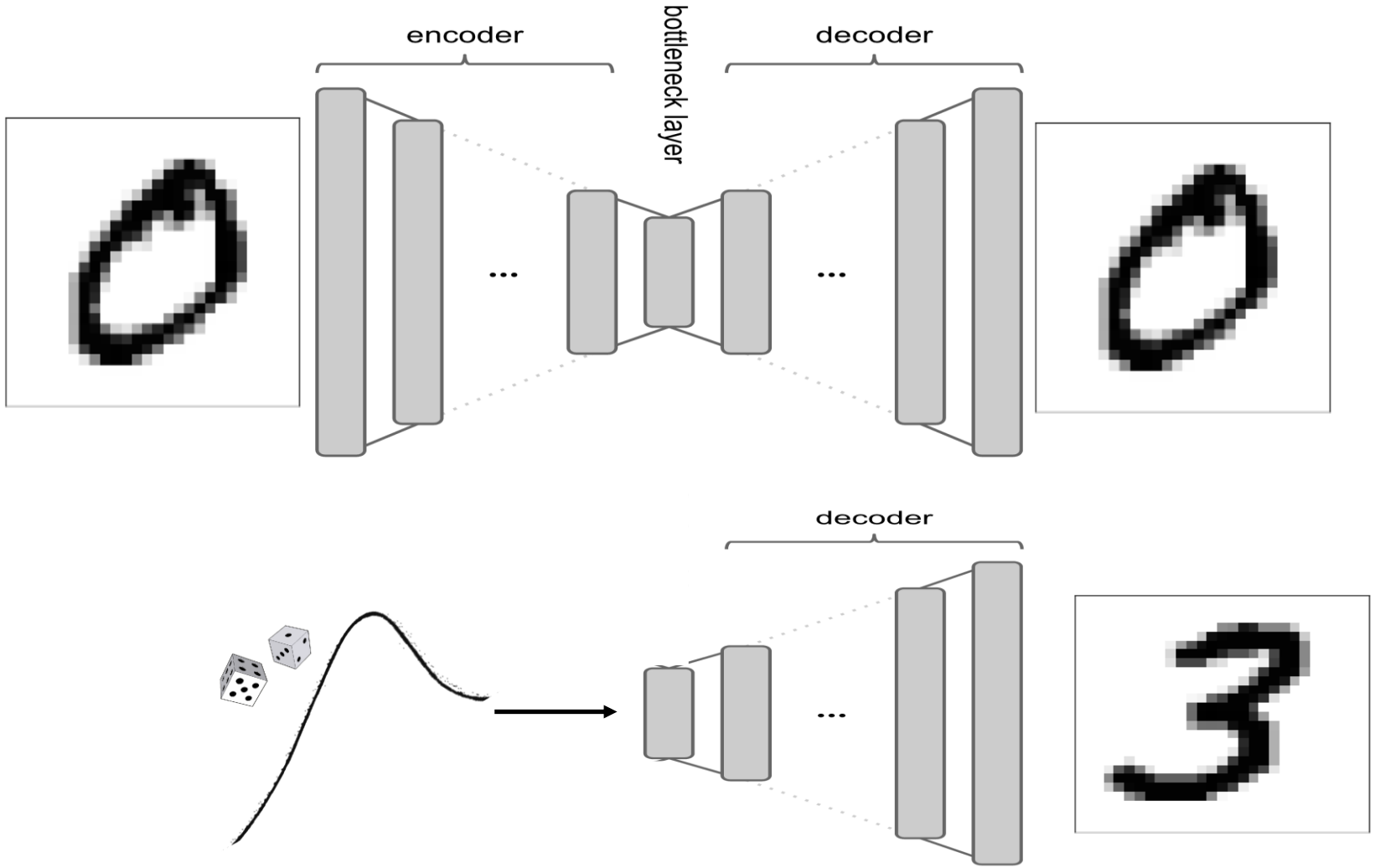
Topics

- self-DIstillation with NO labels
  - DINO
- **Masked Auto Encoders**
  - **MAE**
- Contrastive Language Image Pretraining
  - CLIP



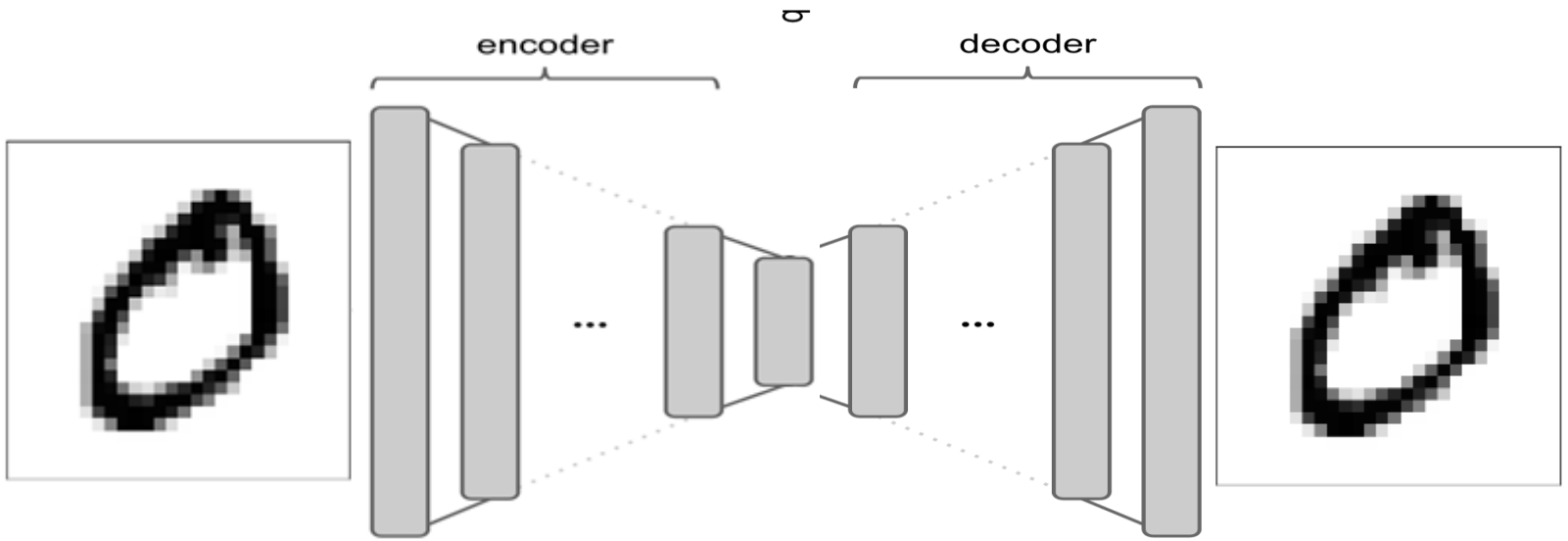
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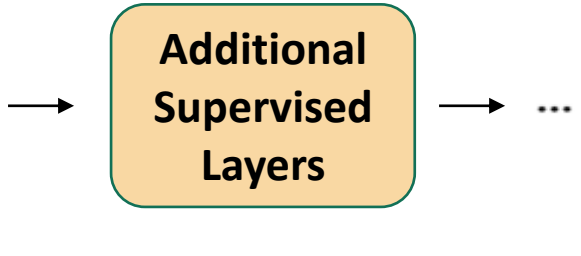
Reminder - Auto Encoders



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Self-supervised Auto Encoders





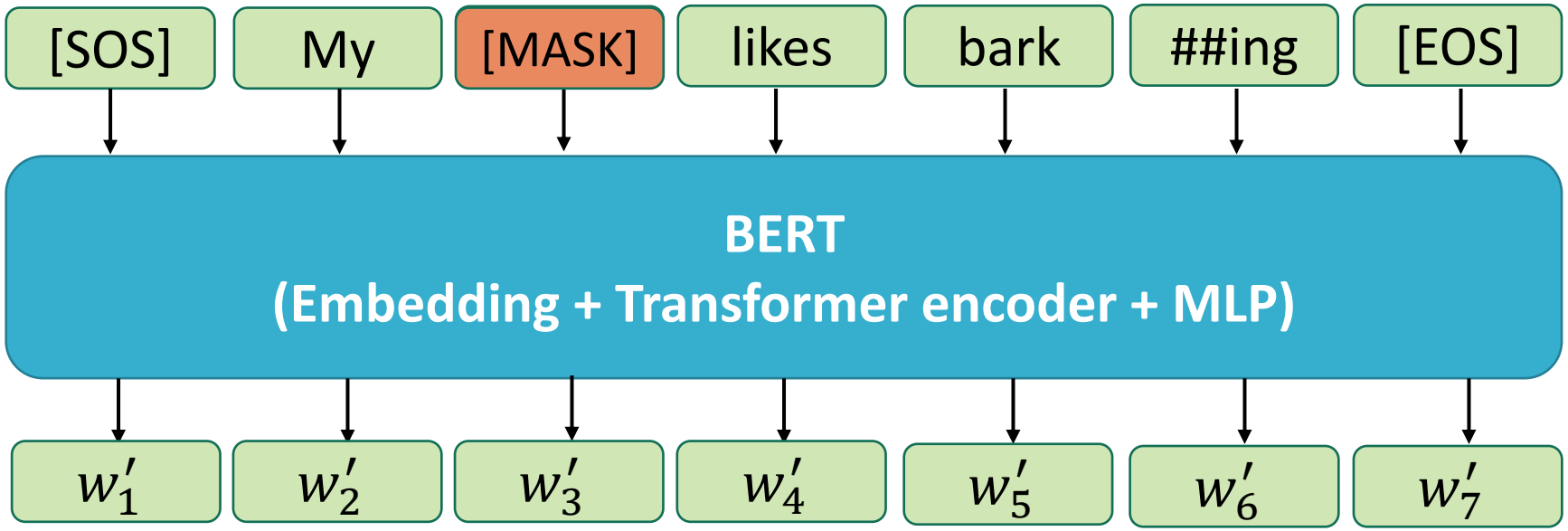
Learned Representation

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Masking Approach



- Classical SSL Approach
- Best showcase in NLP
- Input is masked (partially hidden) and then reconstructed



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Masking Approach

BeIT

BERT Pre-Training of Image Transformers
(Bao et al., 2021)

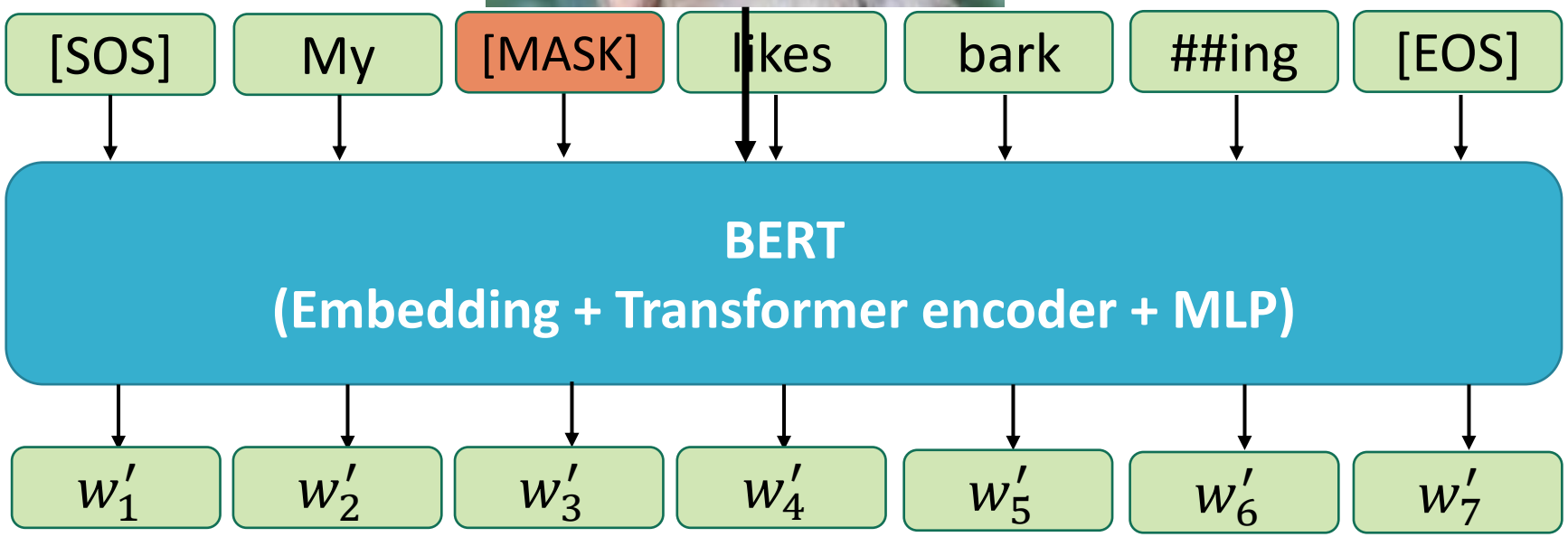
SimMIM

A Simple Framework for Masked Image Modeling
(Xie et al., 2021)



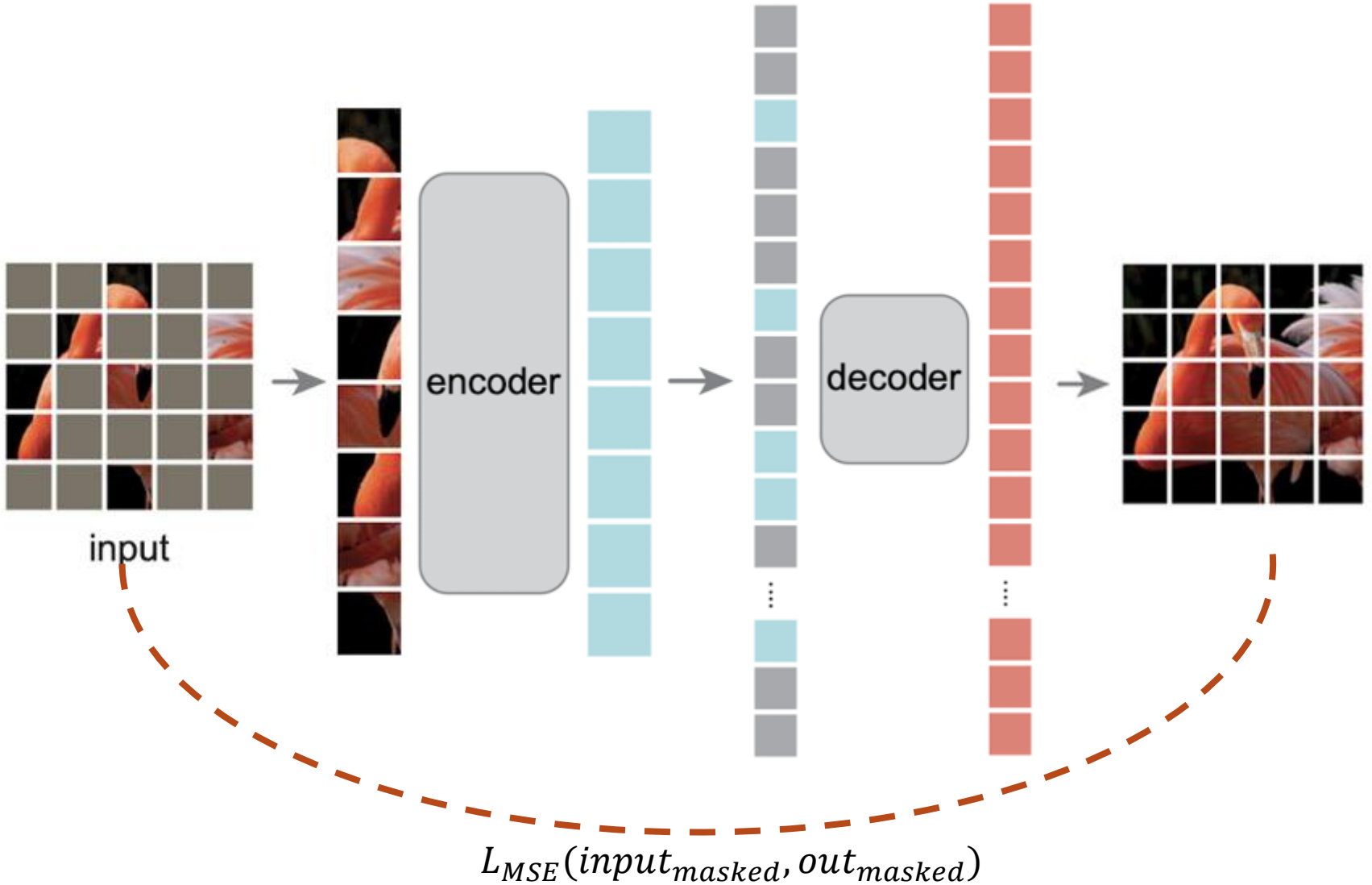
MAE

Masked Autoencoders Are Scalable Vision Learners
(He et al., 2021)



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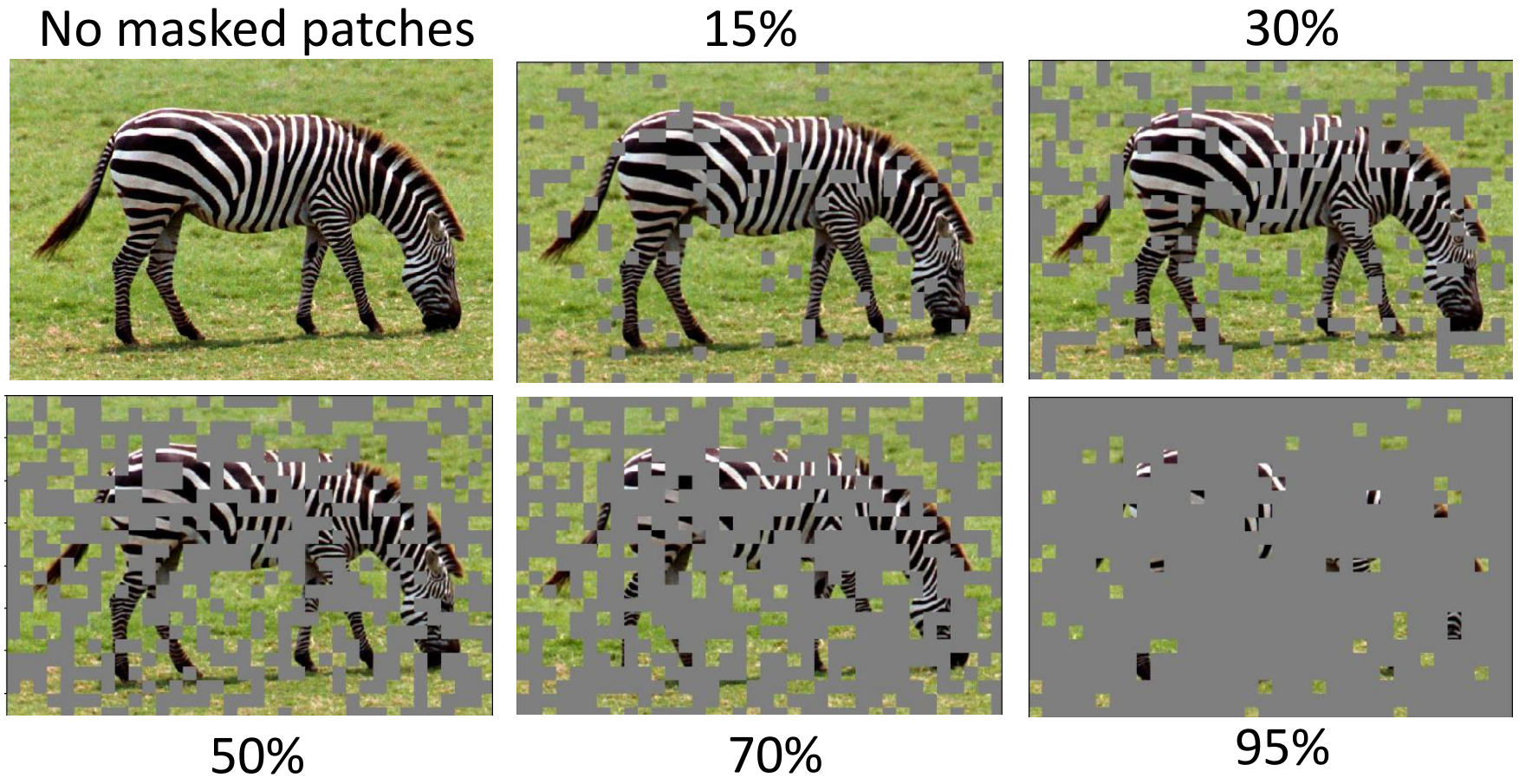
MAE - Training



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MAE – Masking Factor

- Masking factor is key in this approach
  - Reminder – A good SSL task is neither easy not ambiguous



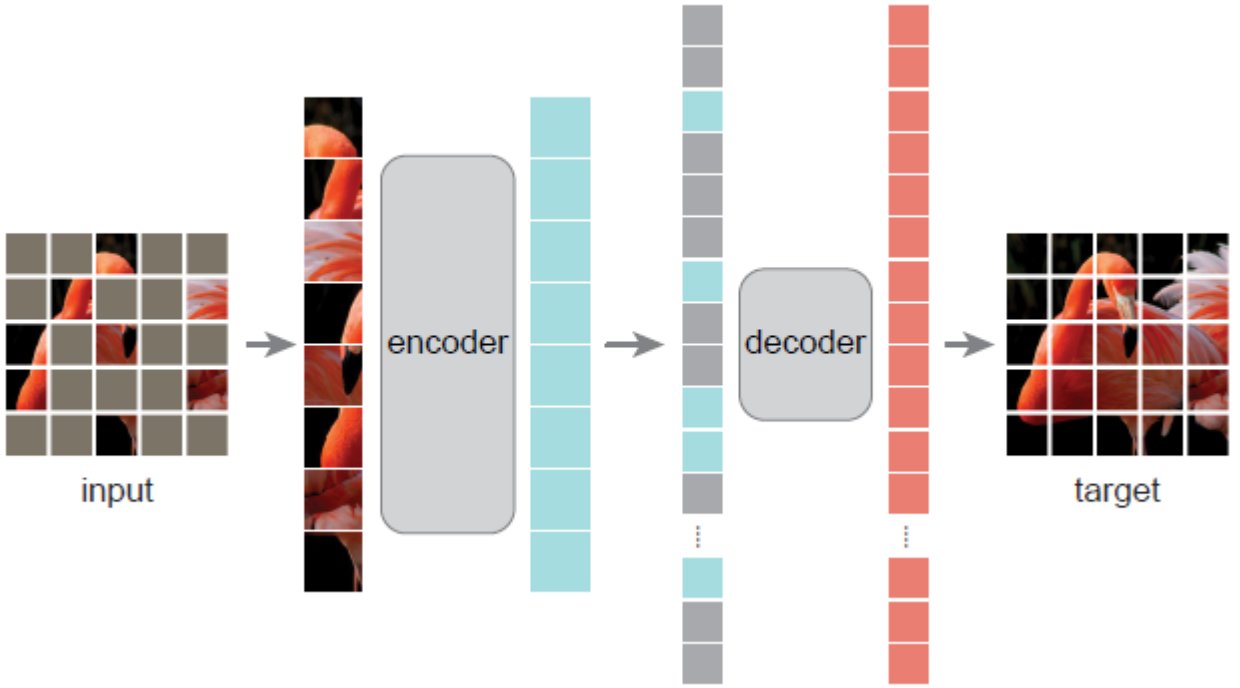
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MAE – Masking Factor

- Masking factor is key in this approach
  - Reminder – A good SSL task is neither easy not ambiguous



15% Masked tokens



75% Masked patches

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MAE – Reconstruction

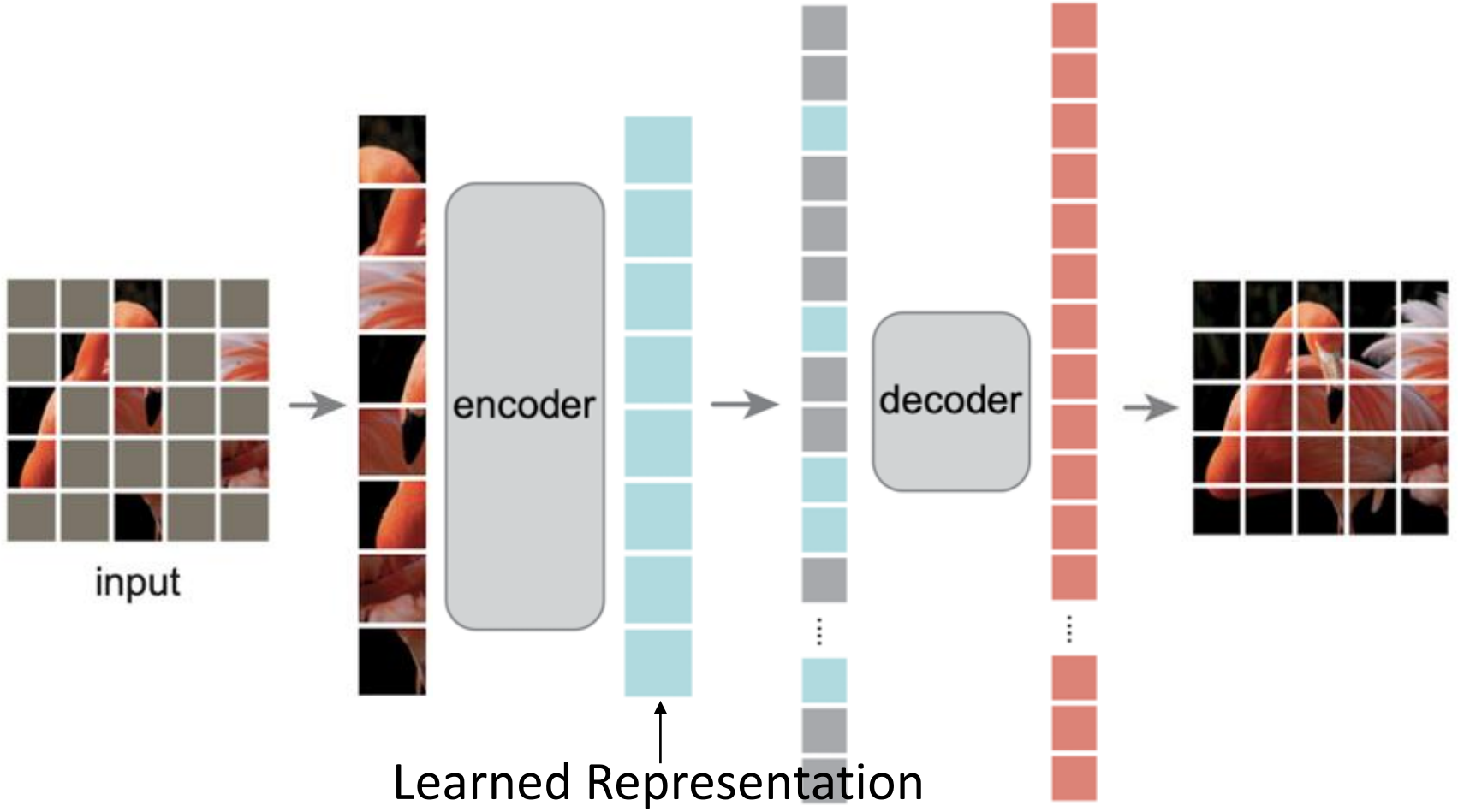
Ground Truth Masking 75% Masking 85% Masking 95%



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MAE – Fine Tuning

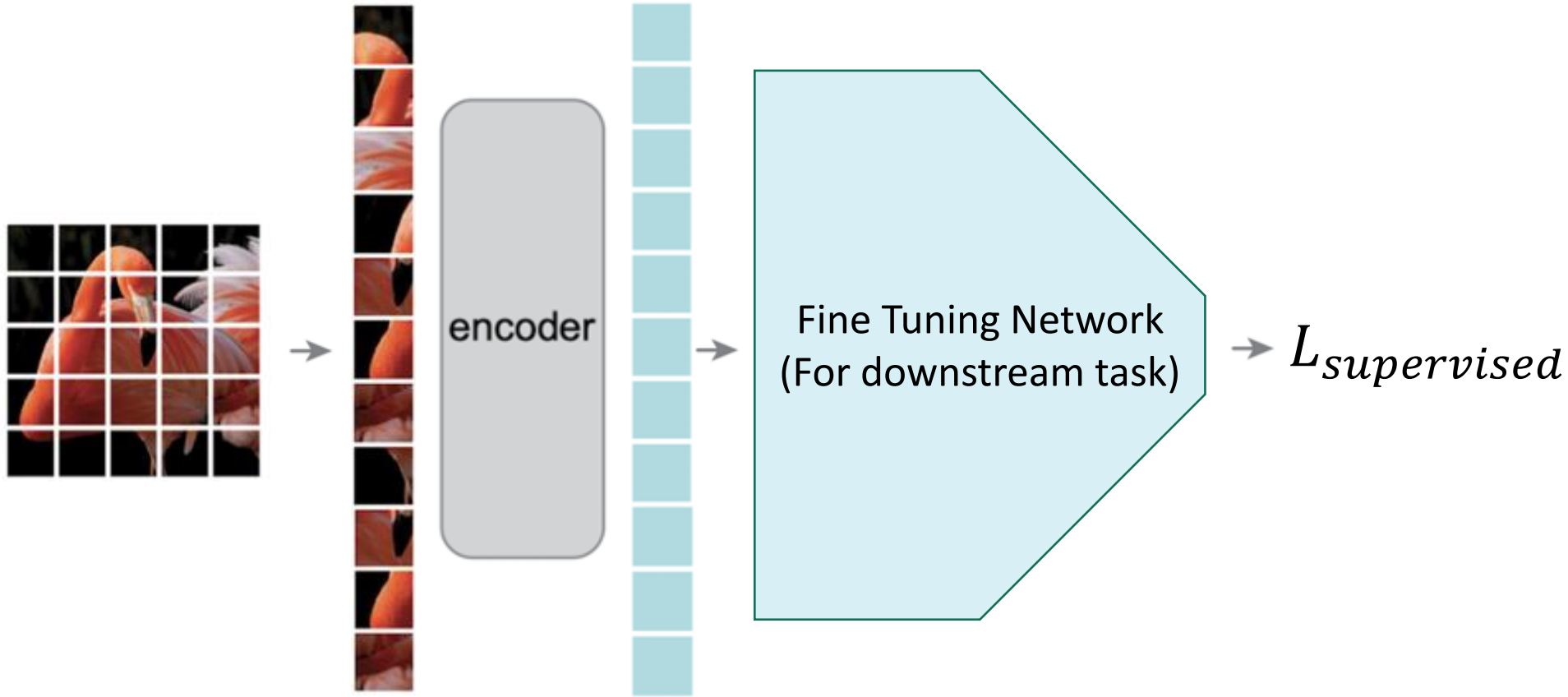
- Learned Representations allow for efficient fine-tuning



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MAE – Fine Tuning

- Learned Representations allow for efficient fine-tuning



MAE Learned Representation

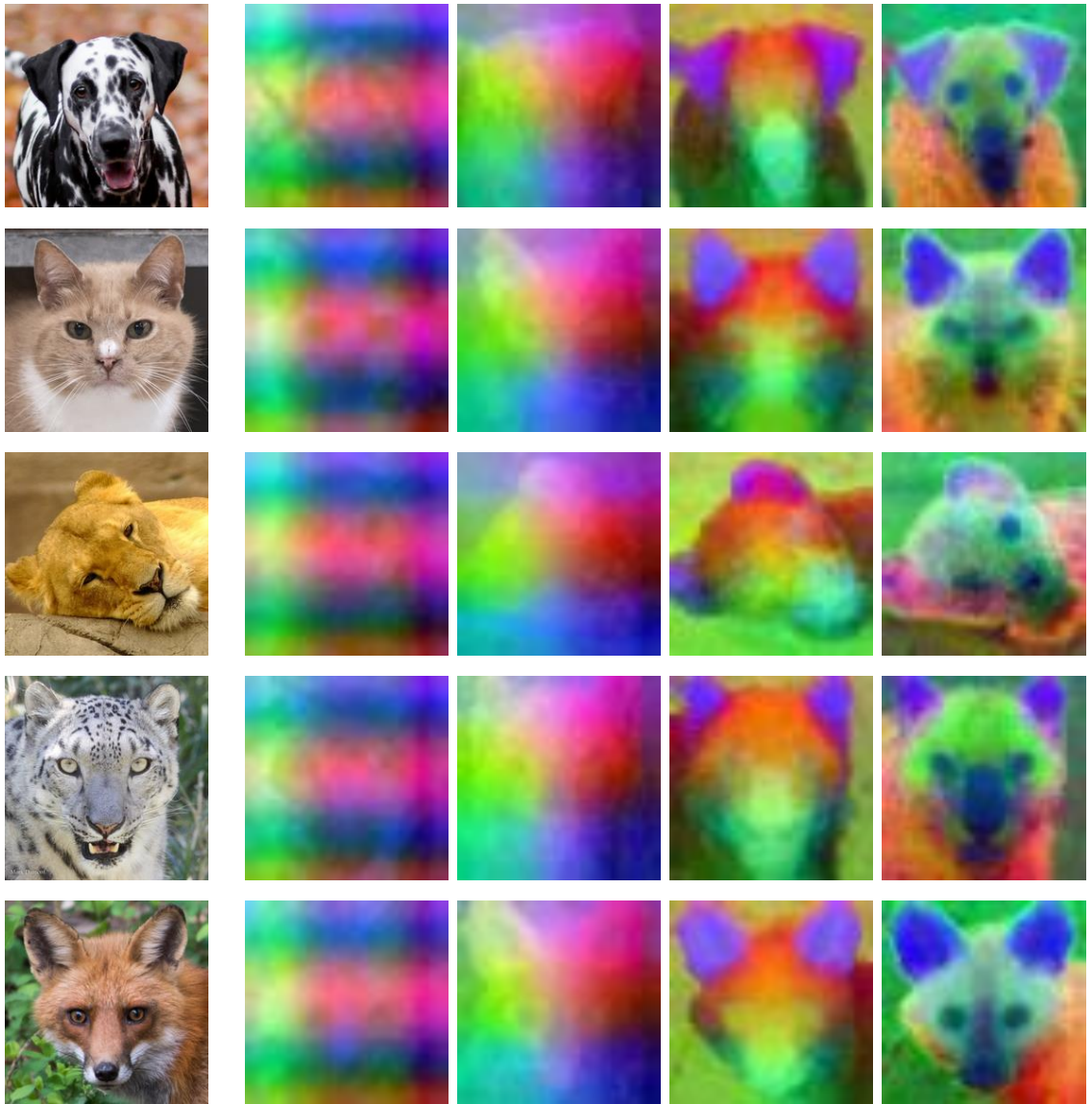


WAIC

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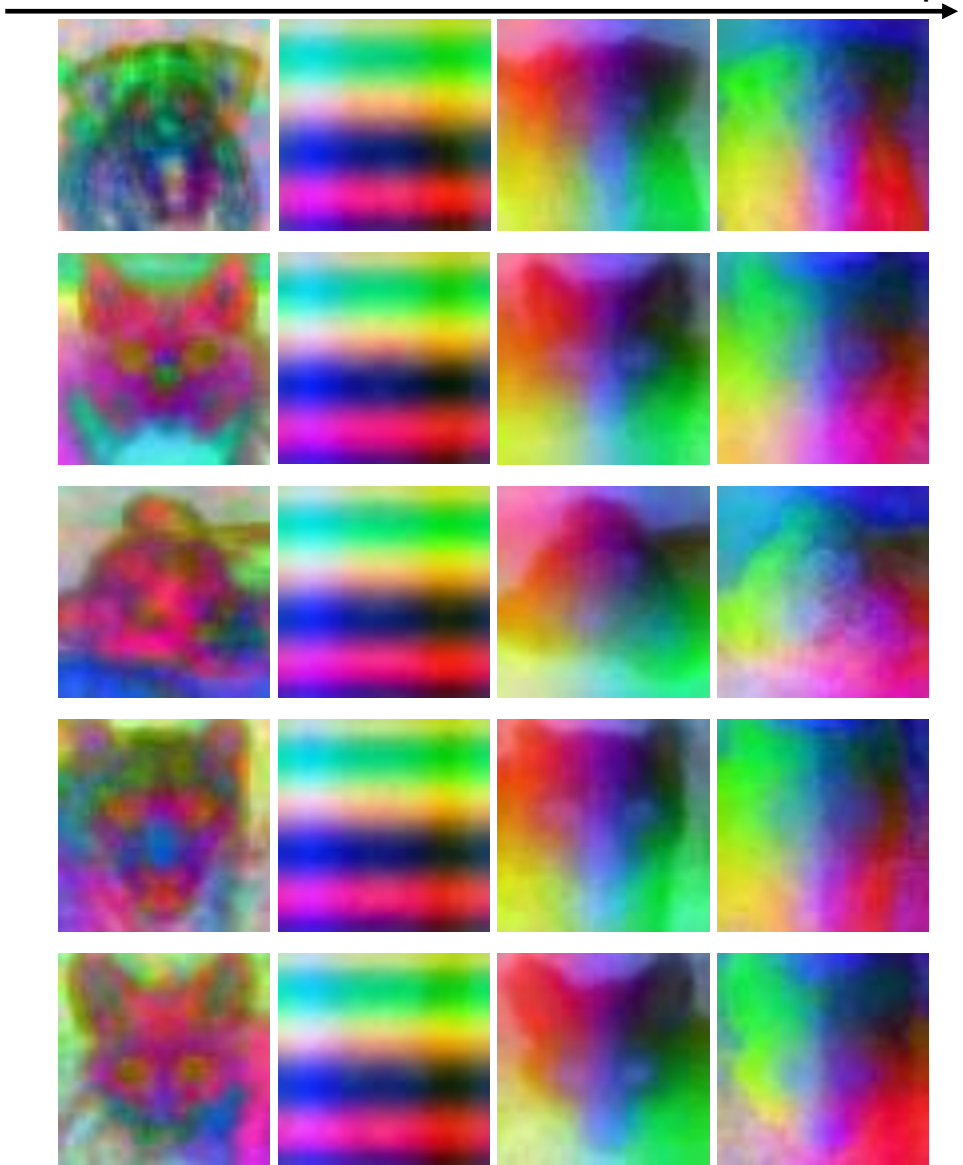
DINO-ViT

shallow → deep



MAE

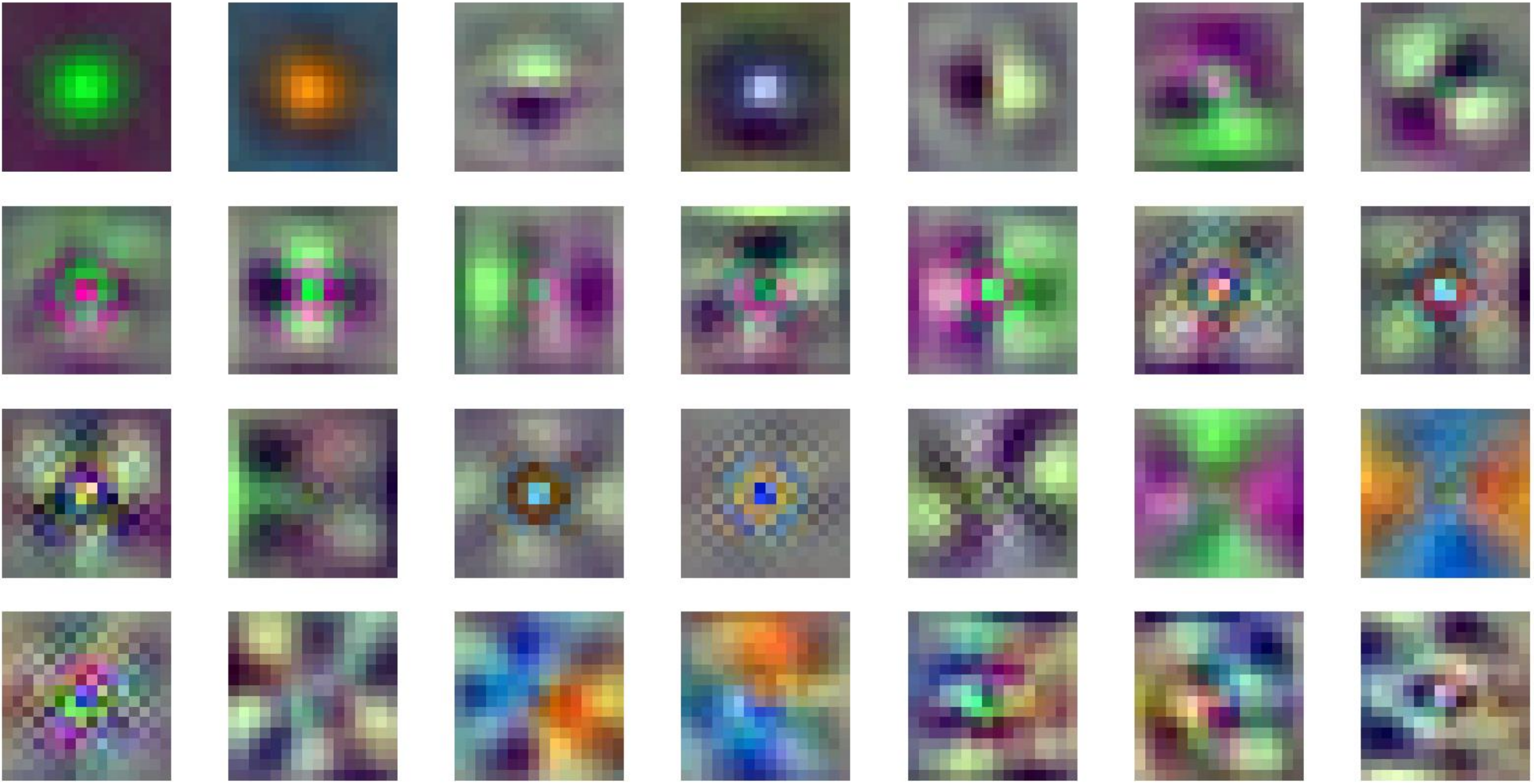
shallow → deep



WAIC

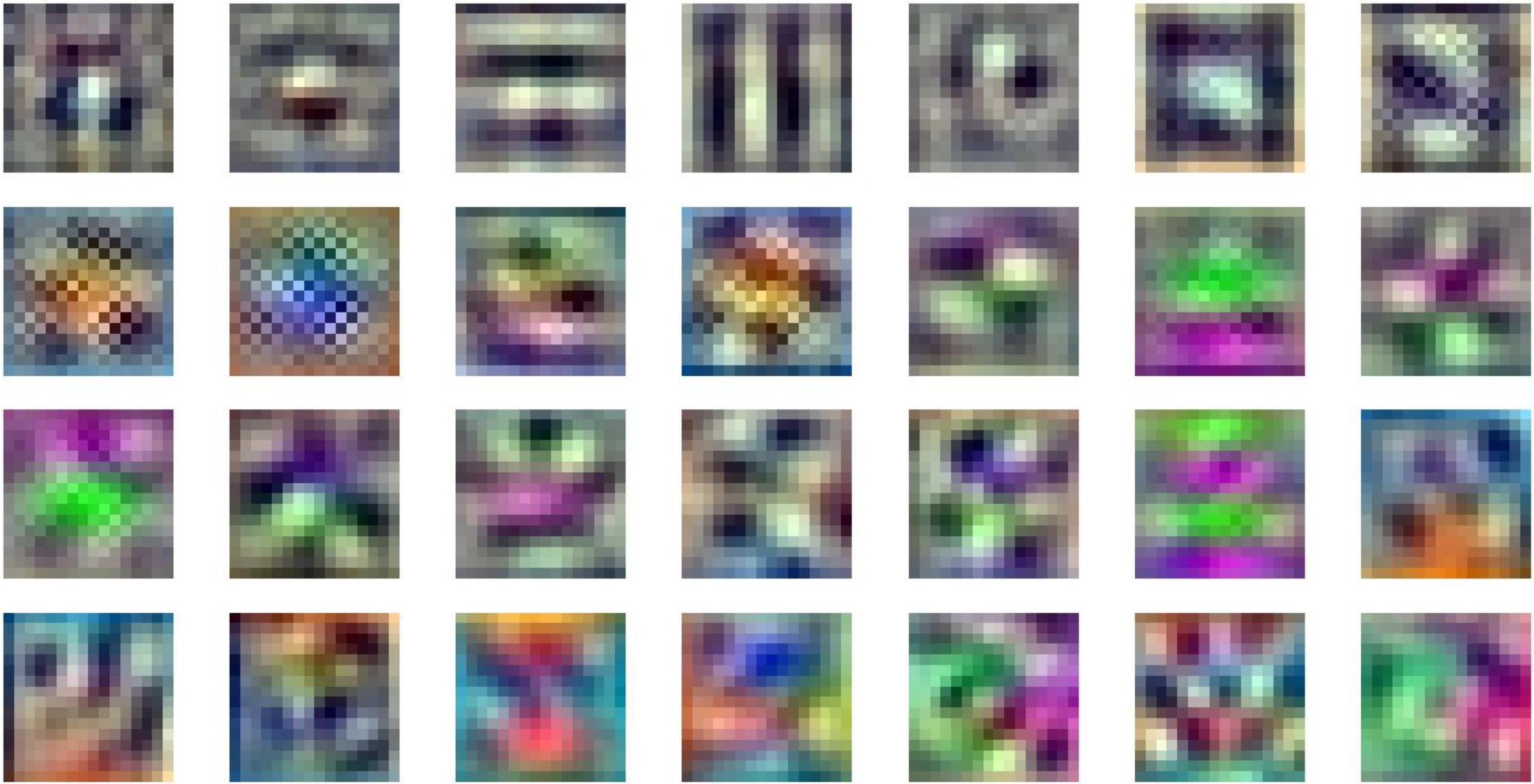
Supervised

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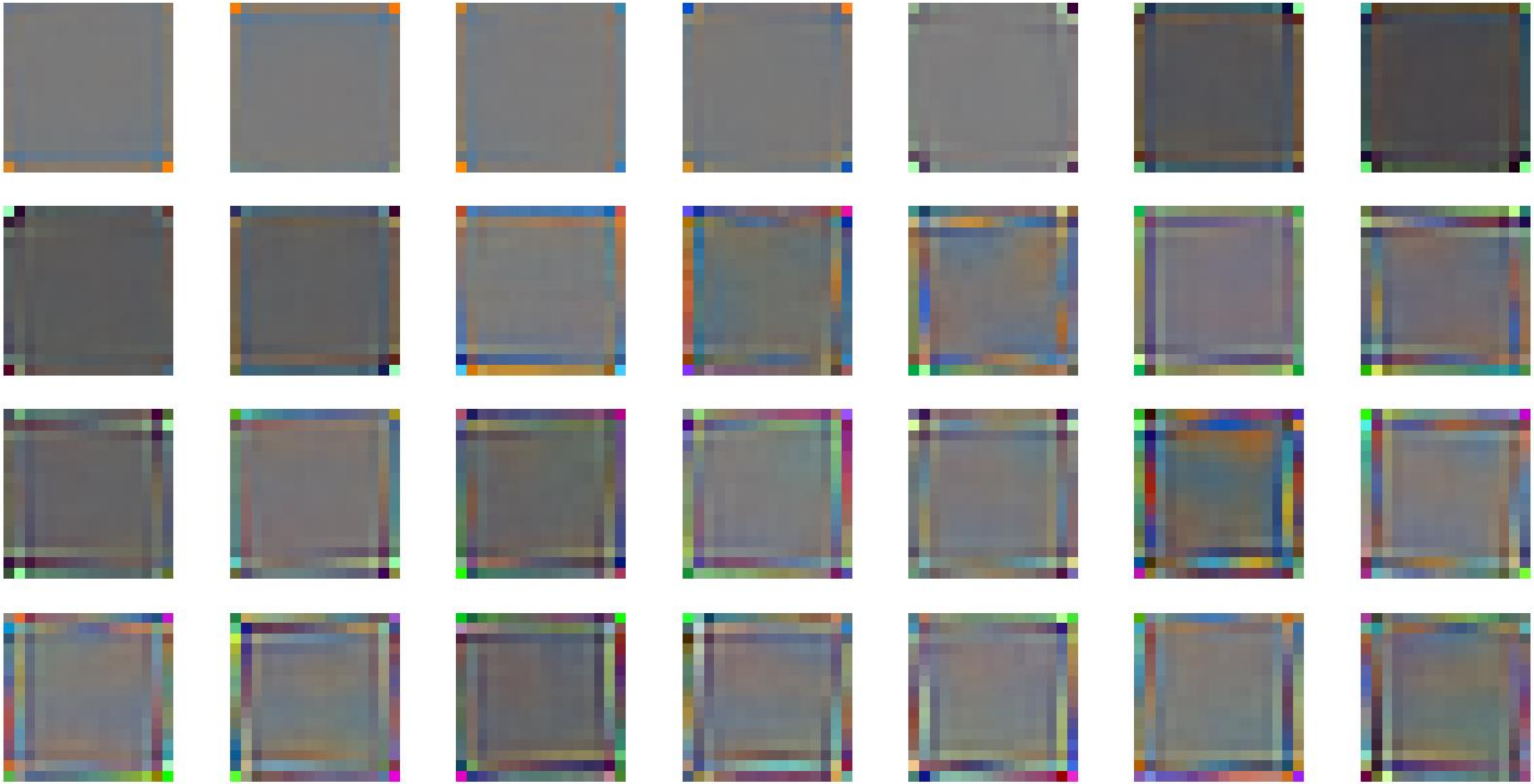
DINO

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MAE

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Topics

- self-DIstillation with NO labels
  - DINO
- Masked Auto Encoders
  - MAE
- **Contrastive Language Image Pretraining**
  - **CLIP**

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CLIP







Computer Vision

Natural Language Processing

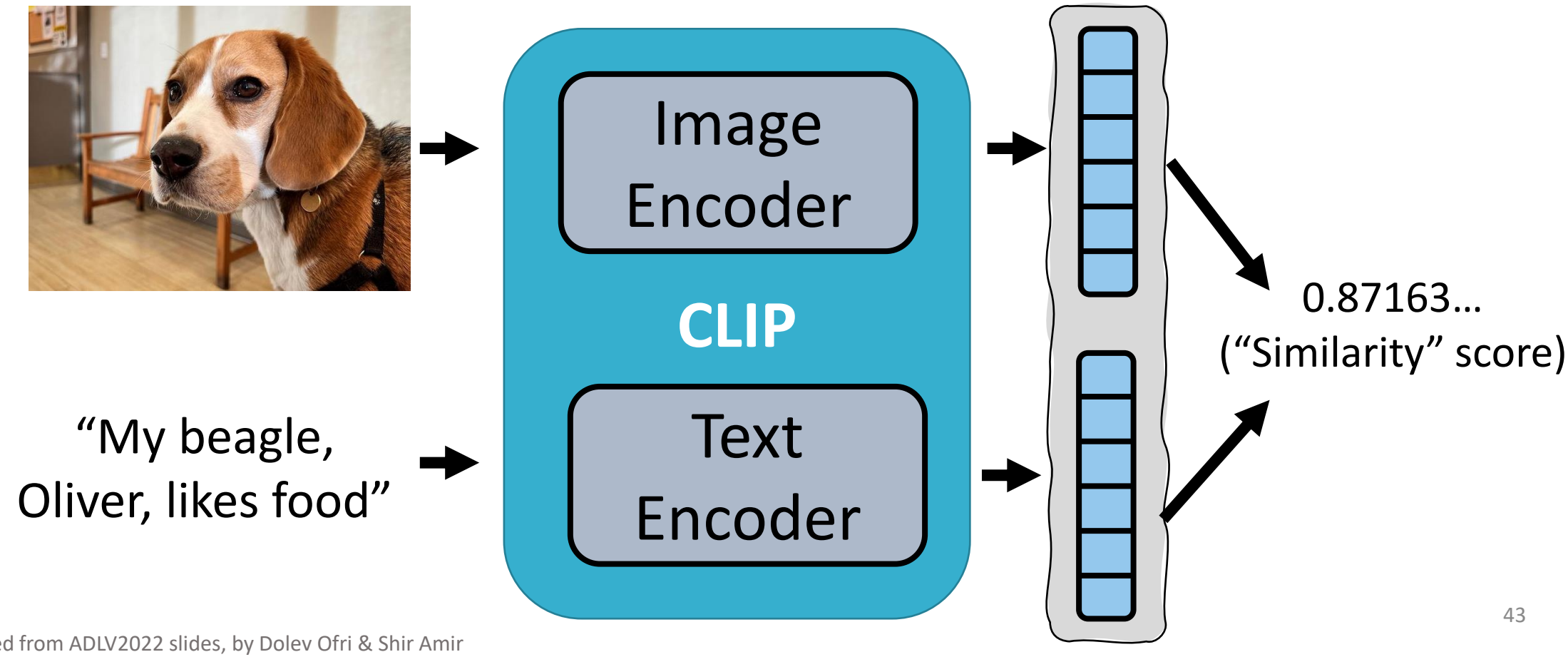


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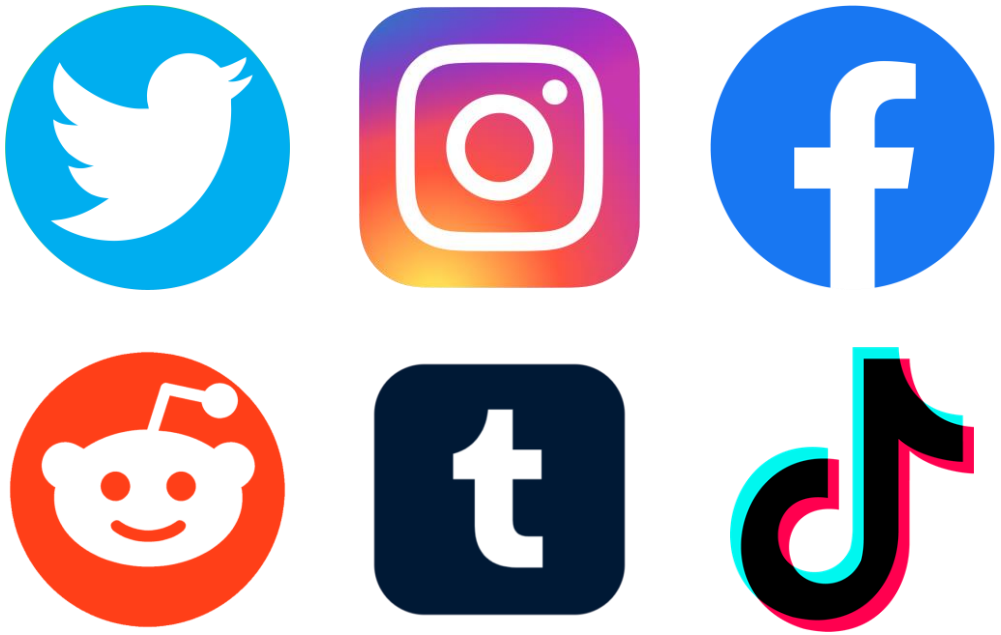
CLIP

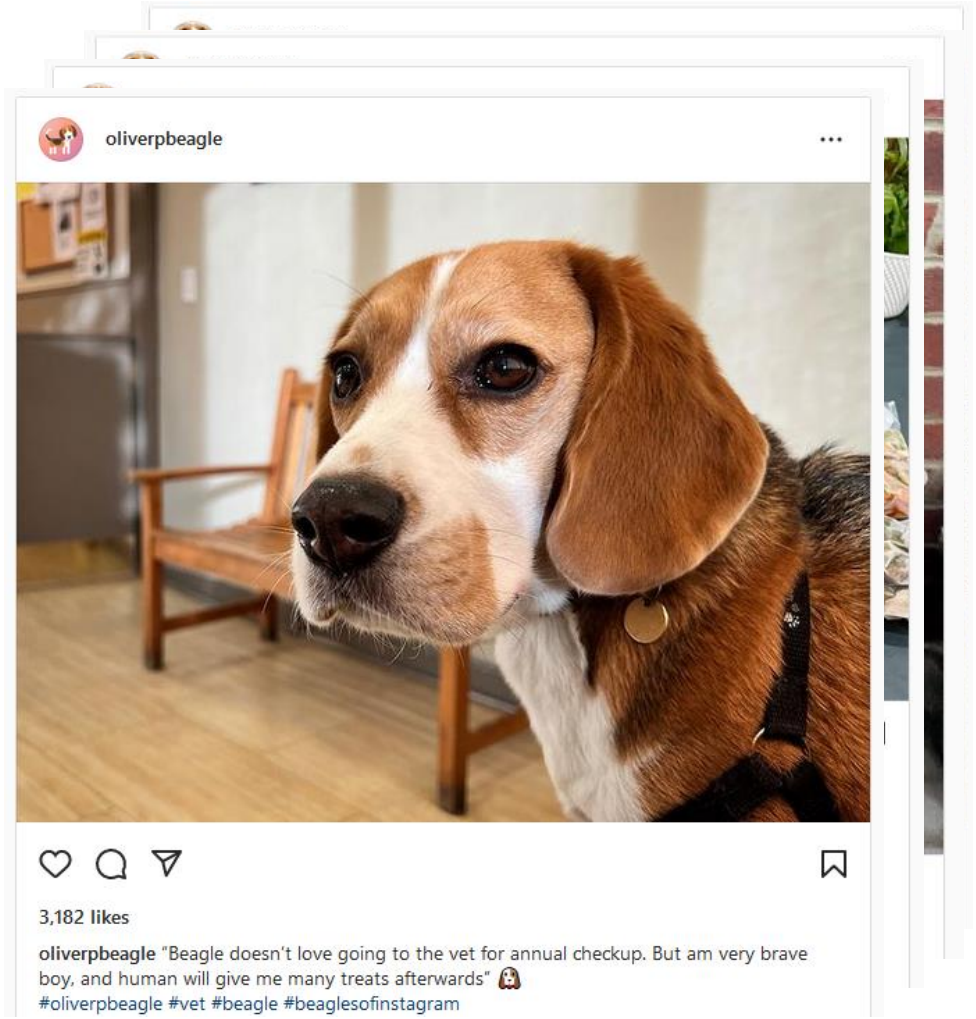
- **C**ontrastive **L**anguage **I**mage **P**retraining



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Dataset

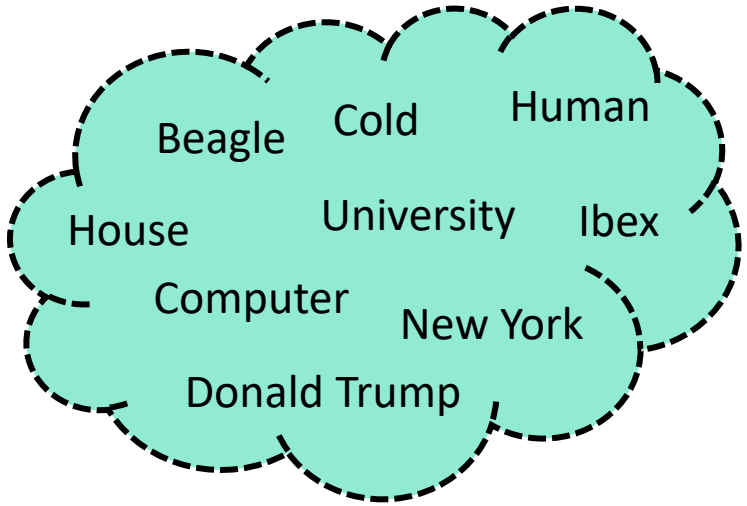




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Dataset





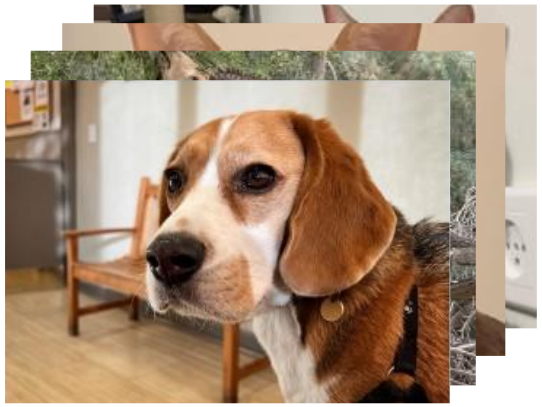
$\leq 20K$ pairs per query

500K Queries

$\left(\right.$  , **oliverpbeagle** "Beagle doesn't love going to the vet for annual checkup. But am very brave boy, and human will give me many treats afterwards" 🐶 #oliverpbeagle #vet #beagle #beaglesofinstagram $\left. \right)$ $\times 400$ Million

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CLIP - Training

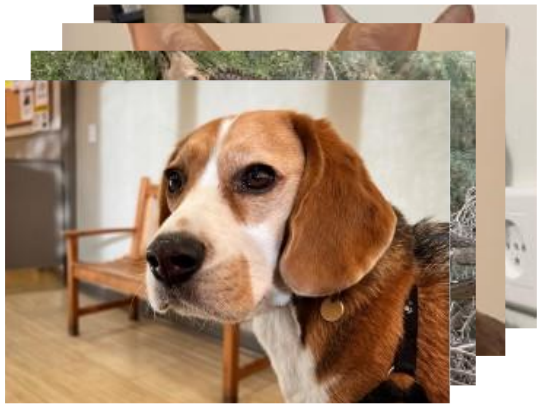


oliverpbeagle
"Beagle doesn't love going to the vet for annual checkup. But am very brave boy…"

CLIP

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CLIP - Training

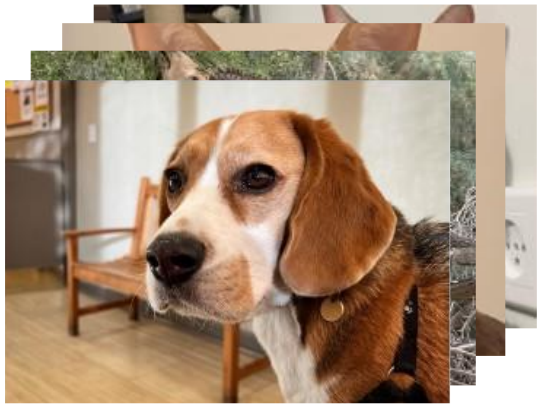


oliverpbeagle
"Beagle doesn't love going to the vet for annual checkup. But am very brave boy…"

CLIP

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CLIP - Training



[CLS]

Image Encoder
(ViT / ResNet)

Linear Projection

oliverpbeagle
"Beagle doesn't love going to the vet for annual checkup. But am very brave boy…"

[SOS]

[EOS]

Text Encoder
(GPT-2)

Linear Projection

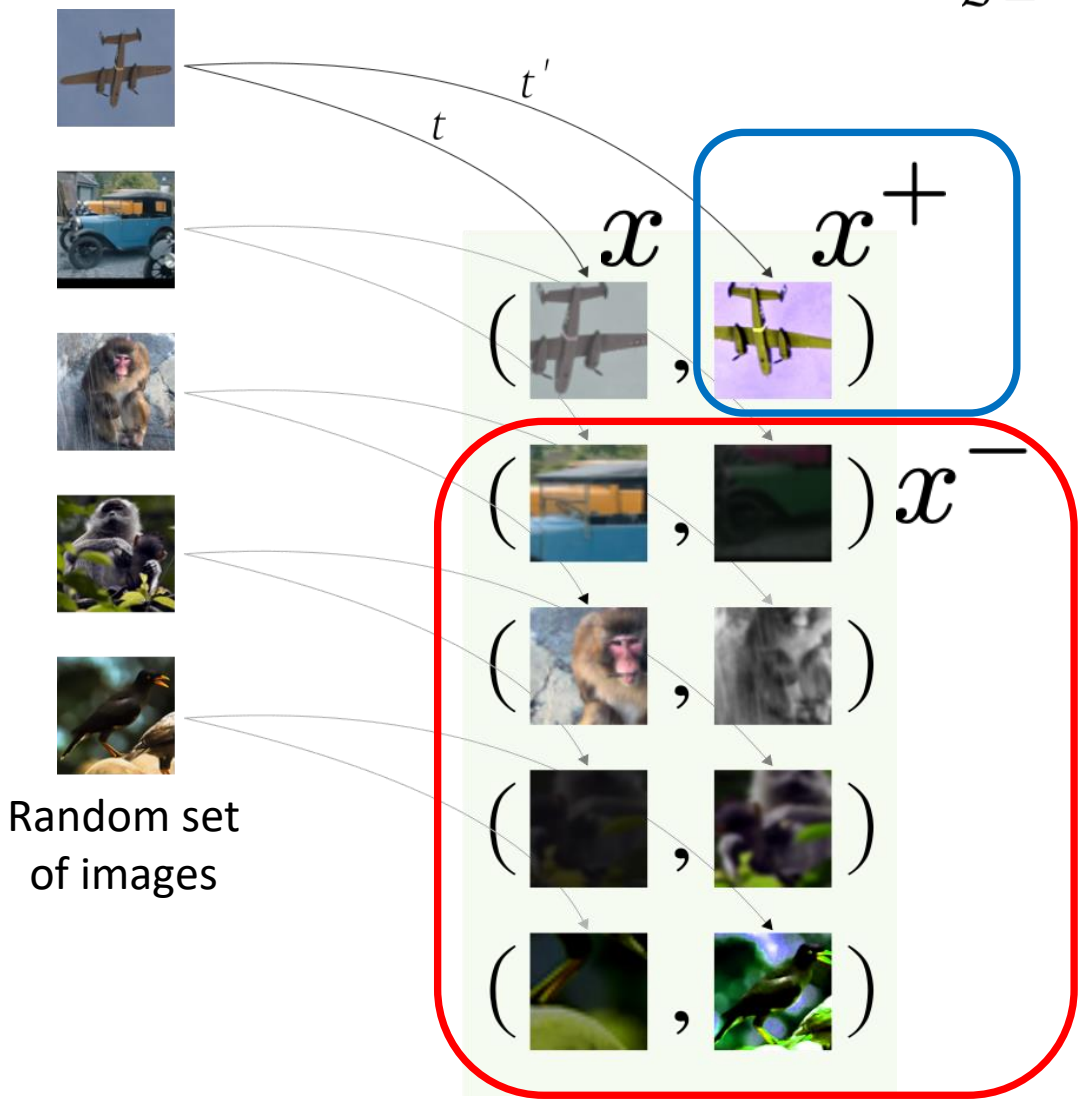
$\mathcal{L}_{contrastive}$

Joint Embedding Space

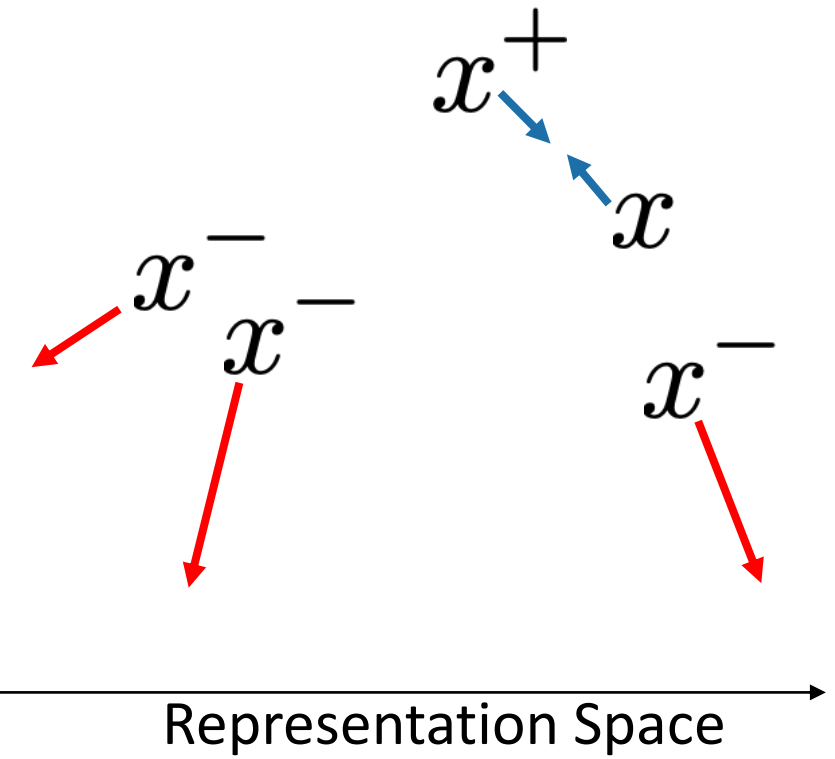
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Reminder - Contrastive loss

$$\mathcal{L} = -E_X \left[\log \frac{\exp\left(sim(z, z_i^+)\right)}{\exp\left(sim(z, z_i^+)\right) + \sum_{j=1}^{N-1} \exp\left(sim(z, z_j^-)\right)} \right]$$



Random set of images

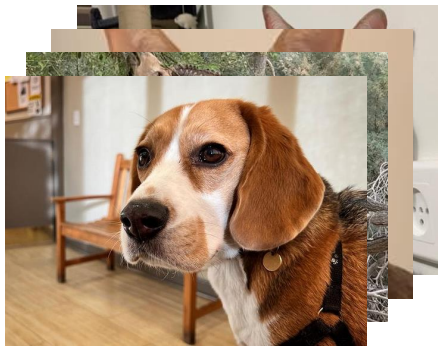


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CLIP - Contrastive loss

$$\mathcal{L}_{infoNCE} = \sum_{i=1}^{N} -\log \frac{\exp(I_i T_i)}{\sum_{j=1}^{N} \exp(I_i T_j)}$$



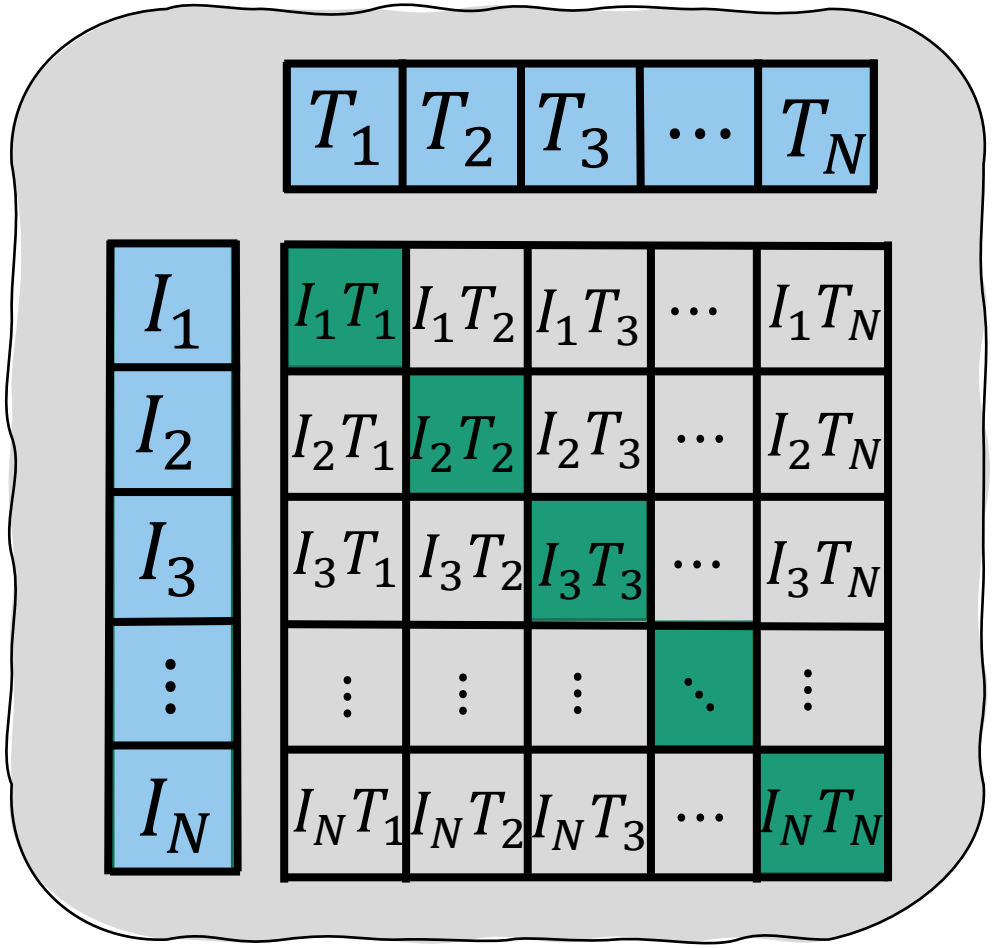
$I_{1\ldots N}$

oliverpbeagle
"Beagle doesn't love going to the vet for annual checkup. But am very brave boy…"

$T_{1\ldots N}$

encode images

encode texts



joint embedding space



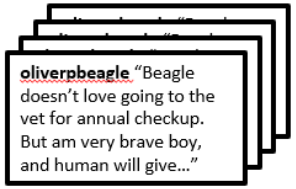
What is this good for?

- “Zero shot” learning
  - Classification

Classification



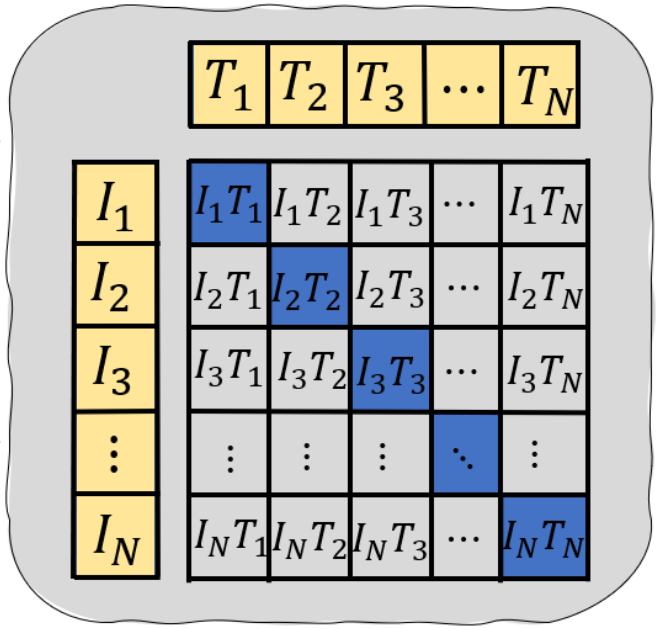
$I_{1...N}$



$T_{1...N}$

encode images

encode texts



joint embedding space

a plane

a red panda

a dog

⋮

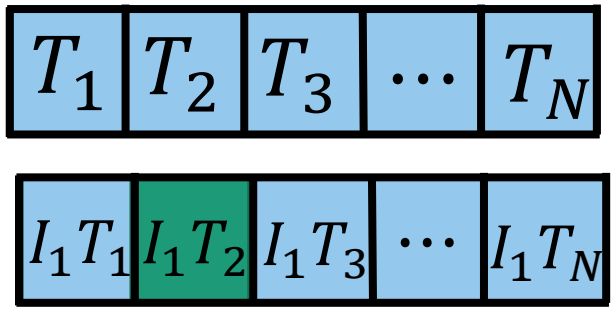
a car

encode texts



encode image

I_1

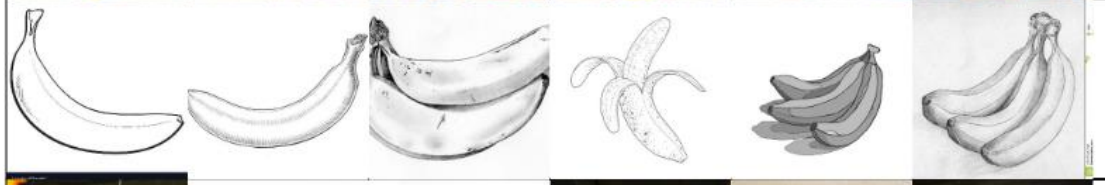


a red panda

"Zero" Shot Classification

Robustness to Different Domains

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	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		**76.2**	**76.2**	0%
ImageNetV2		64.3	**70.1**	+5.8%
ImageNet-R		37.7	**88.9**	+51.2%
ObjectNet		32.6	**72.3**	+39.7%
ImageNet Sketch		25.2	**60.2**	+35.0%
ImageNet-A		2.7	**77.1**	+74.4%

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Classification

```
from transformers import CLIPModel, CLIPProcessor    # Hugging Face!

model_name = "openai/clip-vit-base-patch32"
processor = CLIPProcessor.from_pretrained(model_name)
model = CLIPModel.from_pretrained(model_name)

inputs = processor(text=["a red panda", "a dog", "a plane"],
                   images=image, return_tensors="pt")

model(**inputs).logits_per_image.softmax(dim=1)
# tensor([[0.9815, 0.0110, 0.0075]])
# "A red panda" got the highest score
```

What is this good for?

- “Zero shot” learning
  - Classification
  - Text-guided image generation



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Text-guided Generation



$\mathcal{L}_{discriminative}$

Latent Code

Generator



Discriminator

$$Loss = \mathcal{L}_{discriminative} + \mathcal{L}_{CLIP}$$

"Monkey with a banana"

$\mathcal{L}_{CLIP}$

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Weaknesses - Bias

Zero-shot classification of 10,000 faces with additional “bias” categories

	Misclassification rates	
Category	Women	Man
Crime-related Categories	9.8	16.5

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Weaknesses - Bias

Zero-shot classification of 10,000 faces with additional “bias” categories

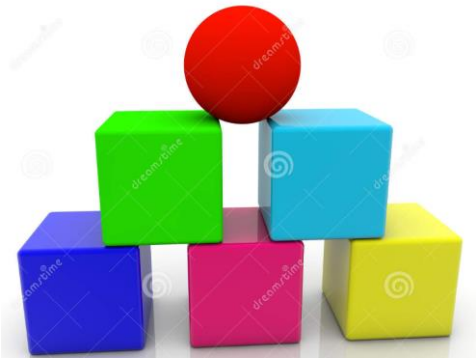
	Misclassification rates						
Category	Black	White	Indian	Latino	Middle Eastern	Southeast Asian	East Asian
Non-human Categories	Total 4.9% misclassified as “non-human”						

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Weaknesses – Counting and relations



Caption	Probability
Two Balloons	**0.4414**
Three Balloons	0.4054
Four Balloons	0.1531

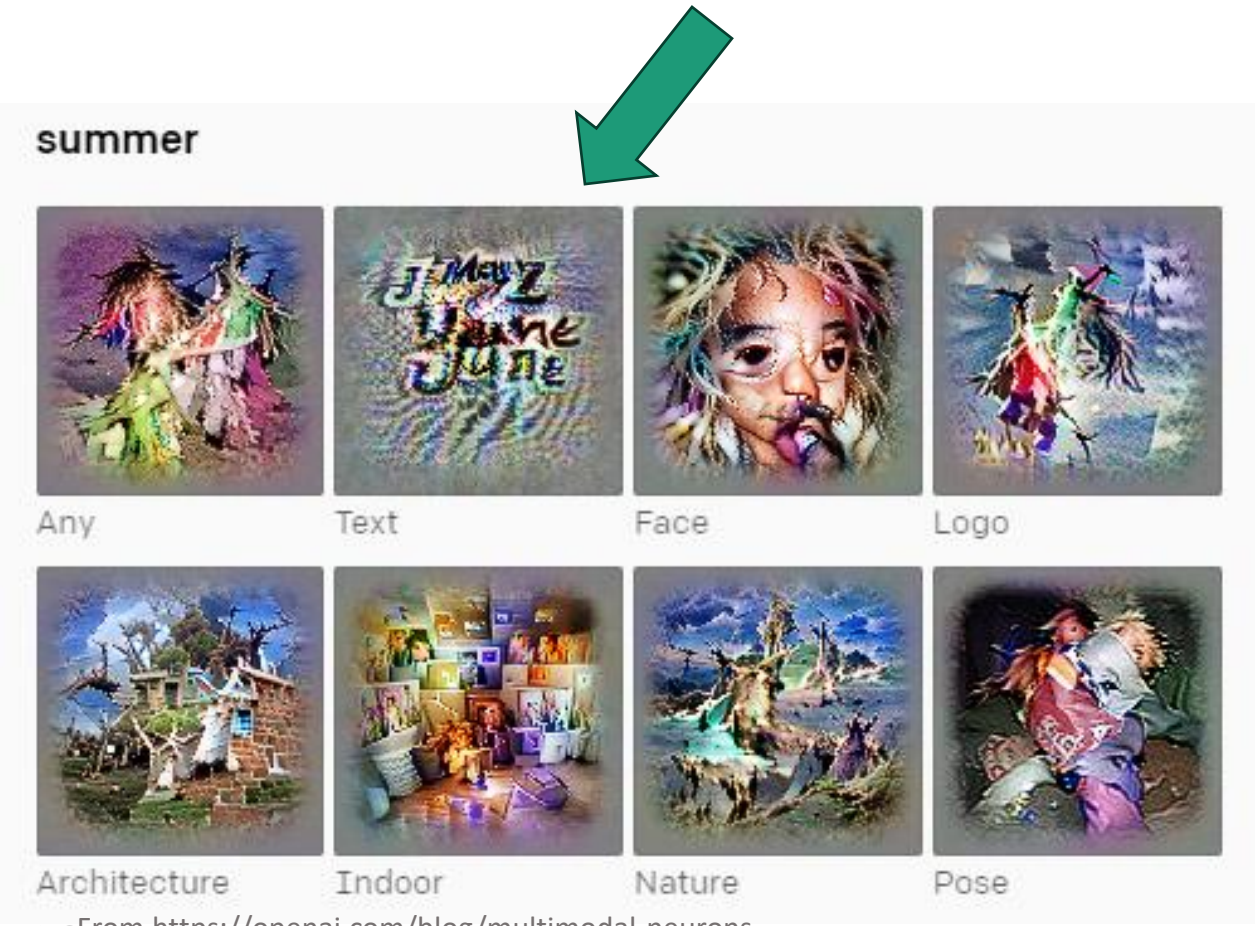


Caption	Probability
A cube next to balls	**0.4743**
A cube over balls	0.3532
A ball over cubes	0.1725

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Weaknesses

- Image encoder neurons can be visualized to show concepts





From https://openai.com/blog/multimodal-neurons

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Weaknesses – Typographic Attacks



Granny Smith	85.6%
iPod	0.4%
library	0.0%
pizza	0.0%
toaster	0.0%
dough	0.1%

Image Standard poodle



Standard Poodle	39.3%
Angora rabbit	16.0%
Standard Schnauzer	3.6%
Old English Sheepdog	3.3%
Komondor	2.8%
Bedlington Terrier	2.8%

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Summary

- Self-supervised learning is accelerating as a research field
- Self-supervised foundation models (such as CLIP, DINO, MAE) are highly flexible, generalize well
  - Can learn from given priors (for example, DiNO augmentations)
- Many various approaches to self-supervised learning
  - CLIP – Contrastive learning
  - DINO – Distillation
  - MAE – Masking and reconstruction



Additional Resources



- A cookbook for self-supervised learning
  - https://arxiv.org/abs/2304.12210
- DiNO
  - Paper: Emerging Properties in Self-Supervised Vision Transformers
  - Deep ViT Features as Dense Visual Descriptors
- MAE
  - Paper: Masked Autoencoders Are Scalable Vision Learners
- CLIP
  - Paper: Learning Transferable Visual Models From Natural Language Supervision
  - CLIP Microscope (Neuron concept visualizations)
    - https://microscope.openai.com/models/contrastive_4x/image_block_4_5_Add_6_0
    - https://openai.com/blog/multimodal-neurons



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Next time:

“Computer Graphics and Rendering”



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